

UALBERTA **T**HESIS **U**SER/**W**RITING **G**UIDE

by

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Abstract

This guide is a comprehensive resource crafted for University of Alberta students tackling their theses with LaTeX. It is designed to simplify the thesis-writing process by offering a detailed walkthrough of a custom LaTeX template tailored to meet the university's formatting requirements. The guide goes beyond mere template usage, providing step-by-step instructions, tips, and best practices for creating a well-structured thesis that adheres to academic standards.

It also delves into advanced customization techniques, allowing users to achieve specific stylistic elements and formatting nuances that suit their individual needs. Alongside these, the guide introduces JabRef, a robust reference management tool, and explains how to seamlessly integrate it with LaTeX to streamline citation management. Whether you are a novice or an experienced LaTeX user, this guide equips you with the knowledge and tools necessary to produce a polished, professional thesis. By following its clear and concise instructions, students can confidently navigate the complexities of thesis writing, ensuring their work is both technically sound and perfectly laid out.

Preface

Writing a thesis is no small task, and as a graduate student at the University of Alberta, I quickly realized just how challenging it can be to meet all the formatting requirements while also producing a document that looks professional. Like many others, I started out using traditional word processors, but it didn't take long before I ran into the usual headaches—crashes, file corruption, and formatting issues that seemed to have a mind of their own.

These frustrations led me to explore L^AT_EX as an alternative. I discovered that L^AT_EX not only provided a way to keep my content and formatting separate but also offered a more reliable and consistent way to produce a high-quality thesis. The learning curve was steep, but once I got the hang of it, I found it to be a game-changer.

The original template document was born out of my experience with L^AT_EX and the desire to make the thesis-writing process a bit less daunting for others who might be in the same boat. The original template was really just a stripped down version of my thesis, with some useful macros and every package I used crammed into a custom `*.cls` file and a bit of folder organization. To be brief, it was pretty cumbersome to use unless you already knew L^AT_EX. My goal here with this new document is to provide a comprehensive guide that not only walks you through the basics of L^AT_EX but also gives you practical examples and best practices¹ to follow.

I built this template and class from the ground up with the aim of reducing the typical L^AT_EX

¹These might not be the “best” practices, however, these are practices that I follow to make my work more constant.

learning curve. I have tried to keep it as simple as possible while still making it powerful enough to handle everything you will need for a thesis. I hope this document makes your life a little easier and that you find L^AT_EX as useful as I have.

Disclaimer: This guide provides practical insights and tips for writing a thesis, based on the information available at the time of writing. However, it remains *your responsibility* to verify the accuracy of all content, conclusions, and interpretations. Official university guidelines, requirements, and advice *always* takes precedence over this guide. I assume *no responsibility* for any potential inaccuracies, misrepresentations, or unintended consequences resulting from the use of any material presented in this guide or the supporting documentation and files.

To my darling partner. Your light guides me through the uncertainty this world has thrown at us.

“Etiam ac leo a risus tristique nonummy. Donec dignissim tincidunt nulla. Vestibulum rhoncus molestie odio. Sed lobortis, justo et pretium lobortis, mauris turpis condimentum augue, nec ultricies nibh arcu pretium enim. Nunc purus neque, placerat id, imperdiet sed, pellentesque nec, nisl. Vestibulum imperdiet neque non sem accumsan laoreet. In hac habitasse platea dictumst. Etiam condimentum facilisis libero. Suspendisse in elit quis nisl aliquam dapibus. Pellentesque auctor sapien. Sed egestas sapien nec lectus. Pellentesque vel dui vel neque bibendum viverra. Aliquam porttitor nisl nec pede. Proin mattis libero vel turpis. Donec rutrum mauris et libero. Proin euismod porta felis. Nam lobortis, metus quis elementum commodo, nunc lectus elementum mauris, eget vulputate ligula tellus eu neque. Vivamus eu dolor.”

- Author of the Quote

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While I would love to acknowledge every individual who has influenced this document, I realize that doing so might mean this document never gets finished.

I want to extend my deepest gratitude to my friends and family for their unwavering support throughout all my accomplishments.

A special thank-you goes out to the developers and other L^AT_EX users who have contributed to the L^AT_EX community. Without their work, this guide wouldn't exist in its current form.

I am also incredibly thankful to everyone who provided feedback and support along the way, including those new to L^AT_EX. Your fresh perspectives were invaluable in motivating the transformation from the previous version of this template into a full-fledged class, template, and this newly added guide to writing a thesis, complete with examples of the most requested elements to include.

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List of Symbols

Constants

ϵ_0	Permittivity of Free Space.	$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$
$\hbar$	Reduced Planck Constant.	$\hbar = 1.055 \times 10^{-34} \text{ Js}$
μ_0	Permeability of Free Space.	$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$
π	Mathematical Constant Pi.	$\pi \approx 3.14159$
R_e	Rankine Number.	$R_e = \frac{Lv\rho}{\mu}$
c	Speed of light in a vacuum.	$299,792,458 \text{ m/s}$
g	Acceleration due to Gravity.	$g = 9.81 \text{ m/s}^2$
h	Planck constant.	$6.62607015E-34 \text{ Js}$
k	Boltzmann Constant.	$k = 1.380649 \times 10^{-23} \text{ J/K}$
R	Gas Constant.	$R = 8.314 \text{ J/(mol}\cdot\text{K)}$

Latin

A	Cross-sectional Area.
a	Acceleration.
D	Diameter.
d	Distance.
E	Young's Modulus.

F	Force.
G	Shear Modulus.
I	Area Moment of Inertia.
K	Elastic Constant.
L	Length.
M	Moment.
m	Mass.
P	Pressure.
T	Temperature.
T	Torque.
t	Thickness.
V	Volume.
v	Velocity.

Greek

α	Primary Angle.
δ	Deflection.
λ	Wavelength.
σ	Normal Stress.
τ	Shear Stress.
ε	Strain.

Abbreviations

AC Armor Class.

AL Adventurers League.

AoE Area of Effect.

AoO Attack of Opportunity.

BAB Base Attack Bonus.

BBEG Big Bad Evil Guy.

CHA Charisma.

CON Constitution.

CoS Curse of Strahd.

CR Challenge Rating.

CRPG Computer Role-Playing Game.

D&D Dungeons & Dragons.

DEX Dexterity.

DM Dungeon Master.

DMG Dungeon Master's Guide.

DMPC Dungeon Master Player Character.

DnD Dungeons & Dragons.

DPR Damage Per Round.

DR Damage Reduction.

ECL Effective Character Level.

HD Hit Dice.

HP Hit Points.

IC In Character.

INT Intelligence.

LA Level Adjustment.

LFG Looking For Group.

LoS Line of Sight.

LR Long Rest.

MM Monster Manual.

NPC Non-Player Character.

OGL Open Game License.

OOO Out Of Character.

PB Proficiency Bonus.

PC Player Character.

PHB Player's Handbook.

PP Passive Perception.

RAI Rules As Intended.

RAW Rules As Written.

RNG Random Number Generator.

RP Roleplaying.

SR Spell Resistance.

SRD System Reference Document.

STR Strength.

THAC0 To Hit Armor Class 0.

TPK Total Party Kill.

TTRPG Tabletop Role-Playing Game.

VTT Virtual Tabletop.

WIS Wisdom.

WotC Wizards of the Coast.

XP Experience Points.

Glossary of Terms

Ability Score One of six numbers (Strength, Dexterity, Constitution, Intelligence, Wisdom, Charisma) that represent a character's physical and mental attributes.

AC (Armor Class) A number representing how difficult it is to hit a character in combat.

Advantage/Disadvantage A mechanic where a player rolls two d20s and takes the higher (advantage) or lower (disadvantage) result.

Alignment A character's ethical and moral perspective, such as Lawful Good or Chaotic Evil.

Arcane A type of magic derived from study, such as wizardry.

Backstory The history and background of a character before the campaign begins.

Bonus Action An additional action a character can take during their turn, often granted by class features or spells.

Cantrip A spell that can be cast at will without using a spell slot.

Chaotic Free-spirited and sometimes unpredictable. Can also be reckless or reactionary.

Charisma (CHR) Social skills and sometimes physical appearance.

Class The primary archetype of a character, such as Fighter, Wizard, or Rogue, which determines abilities and progression.

Combat A structured sequence where characters and enemies take turns performing actions like attacking or casting spells.

Concentration A mechanic where certain spells require ongoing focus, and taking damage can force a concentration check to maintain the spell.

Constitution (CON) Physical resilience. This affects hit points and some physical resistances.

d20 A 20-sided die, the primary die used in D&D for most rolls.

Damage Types One of the thirteen (13) categories of damage: acid, bludgeoning, cold, fire, force, lightning, necrotic, piercing, poison, psychic, radiant, slashing, and thunder.

Dexterity (DEX) Agility and accuracy. This affects ranged attacks and dodging.

Dungeon Master (DM) The person who runs the game, narrates the story, and controls the world and NPCs.

Encounter Any situation where players must overcome a challenge, such as combat, puzzles, or social interaction.

Equipment The gear and items a character carries, including weapons, armor, and adventuring tools.

Evil Wicked and often selfish or oppressive.

Experience Points (XP) Points gained from overcoming challenges, used to level up a character.

Familiar A magical creature that assists a spellcaster, often summoned by the spell *Find Familiar*.

Feat A special ability or skill a character can choose instead of an ability score increase at certain levels.

Flanking A tactical position where a character attacks an enemy from the opposite side of an ally, often granting a combat advantage (this rule is optional and varies by DM).

Good Having a respect for life, altruism, and selflessness.

Grapple A combat action where a character attempts to grab and restrain an opponent.

Group Check A mechanic where the success of the party depends on the number of successful rolls among the group.

Hit Points (HP) A measure of a character's health, reduced when taking damage.

Hit Dice (HD) Dice used to determine a character's hit points at each level and for healing during short rests.

Initiative A roll made at the start of combat to determine the order of turns..

Inspiration A DM-awarded bonus that allows a player to gain advantage on a roll.

Intelligence (INT) The ability to process problems and wield certain magic. INT affects the number of skill points received.

Ki A resource used by monks to perform special abilities.

Lawful Abides by a core morality or honor system. Can also be judgmental and close-minded.

Level A measure of a character's progression, determining access to new abilities, spells, and increased hit points.

Long Rest A period of downtime (usually 8 hours) where characters recover hit points and spell slots.

Melee Combat at close range, typically involving hand-to-hand or short-ranged weapons.

Metagaming Using out-of-game knowledge within the game, often discouraged as it can break immersion.

Multiclassing The practice of taking levels in more than one class, allowing a character to gain abilities from multiple classes.

Neutral A balance between Lawful & Chaotic or Good & Evil.

NPC (Non-Player Character) Characters controlled by the DM that players interact with, such as villagers, shopkeepers, or enemies.

Opportunity Attack A reaction that allows a character to make a melee attack against a creature that moves out of their reach.

Party The group of player characters (PCs) adventuring together.

Perception A skill representing a character's ability to notice hidden things, typically rolled as a Wisdom check.

Proficiency Bonus A bonus added to rolls where a character has proficiency, such as with certain skills, weapons, or saving throws.

Quiver A container for holding arrows or bolts, typically used by archers and ranged combatants.

Ranged Attack An attack made with a ranged weapon or spell, targeting an enemy at a distance.

Reaction An instant response to a trigger, such as casting *Counterspell* or making an opportunity attack.

Saving Throw A roll made to resist a spell, trap, or other effect.

Short Rest A brief period of downtime (usually 1 hour) where characters can spend Hit Dice to recover hit points.

Skill Check A roll made to determine the outcome of an action related to a skill, such as Stealth or Acrobatics.

Spell Slot A resource that determines how many spells a character can cast at each level.

Strength (STR) The character's physical strength. This effects the potency of melee attacks.

Turn A player's time to act during a round of combat, typically consisting of movement, an action, and possibly a bonus action or reaction.

Vision Types Various levels of sight in D&D, such as Darkvision, Blindsight, and Truesight.

Weapon Proficiency Determines which weapons a character can use effectively, adding their proficiency bonus to attack rolls.

Wisdom An ability score representing a player character's insight, perception, and willpower.

Wisdom (WIS) Common sense and spirituality.

XP (Experience Points) See Experience Points.

Chapter 1

Introduction

1.1 Background

As a graduate student from the University of Alberta, I am familiar with the challenging task of writing a thesis that adheres to the Faculty of Graduate and Post-Doctorate Studies Minimum Thesis Formatting Requirements. Using a traditional word processor to create a long document filled with equations and figures can be frustrating due to frequent crashes, file $\text{c}\theta\text{r}_{\mu}\text{p}\text{t}\text{l}\text{n}$, unpredictable **formatting** changes, and the inability to output a document in the required PDF/A format for submission to GPS.

To overcome these issues, many students turn to or are recommended L^AT_EX as an alternative to conventional word processors.

L^AT_EX attempts to separate the formatting from the content of the writing. Thus allowing students and researchers to focus on the content and, typically, have the formatting of their documents managed automatically by templates or class and style files. Because the content is independent of the formatting, documents can be written in lightweight text editors or L^AT_EX editors, which also facilitate the compilation of the documents. These editors can often save work after every keystroke and, due to the plaintext format, they are less prone to file corruption.

Moreover, L^AT_EX ensures a consistent and professional appearance throughout the document.

1.2 Objectives

The main objectives of this guide are:

1. To remind you of the current¹ requirement for writing a thesis at the University of Alberta.
2. To assist students and researchers in mastering the nuances of L^AT_EX document preparation.
3. To provide a comprehensive guide on writing a thesis using L^AT_EX and this template.
4. To showcase best practices and examples for structuring and formatting a thesis in L^AT_EX with this template.

1.3 Scope and Limitations

Although there are existing templates for writing a thesis in L^AT_EX for the University of Alberta (see [Table 1.1](#) for a list of all available templates I was able to find), none seem to provide all the necessary information for creating an outstanding thesis. Most templates apply “band-aid” solutions to existing classes, such as `report` or `book`, offering a customized title page and methods for including prefatory pages. However, these templates often fall short by not providing tips and best practices on how to include the various sections and parts that make up a thesis. They also fail to offer a solid foundation for those who are new to L^AT_EX. Because the `report` or `book` classes were not designed for writing a thesis for the University of Alberta, some of these templates involve extensive patching and fixing, resulting in a large *preamble* section at the beginning of the template that can be confusing to new L^AT_EX users and add to the already steep learning curve.

¹Always double check the information presented in this guide against the information provided by the Faculty of Graduate and Post-Doctorate Studies.

Developer	Last Updated	Link to Template Source
Shivam Garg	May 29, 2023	https://github.com/svmgrg/ualberta_thesis_template
Henry Brausen	Feb 11, 2022	https://github.com/henrybrausen/thesis_template
Bernard Llanos	Oct 05, 2019	https://drive.google.com/file/d/1wKS8fu5e6qiVDRt0VUzEtIW8p7uMyz1T/view?usp=sharing
John Bowman	Sep 30, 2019	https://github.com/vectorgraphics/uofathesis
Hongtao Yang & Benjamin Bernard GAME	Sep 28, 2017	https://github.com/adrs0049/ThesisTemplate
& Hongtao Yang	Feb 03, 2016	https://www.ualberta.ca/computing-science/media-library/grad/candidacy-template-tex.tex
Steven Taschuk	Mar 21, 2012	https://github.com/stebulus/ualberta-math-stat-templates/tree/master/thesis
CMENG	Jul 19, 1999	https://sites.ualberta.ca/CMENG/research/new-control/stythes.html

Table 1.1: List of Other Available Templates.

This template (`ualberta.tex`), document class (`ualberta.cls`), and guide (`guide.pdf`) aim to address these shortcomings by providing all the necessary information to create a well-structured thesis, along with examples to assist in formatting your thesis written in L^AT_EX. To ensure the robustness and ease of maintenance, I developed the class file from the ground up keeping the additional required packages to a minimum. This makes this L^AT_EX solution easier to maintain, update, and customize to suit different needs from different areas of the University of Alberta. A key goal with this was to reduce the traditionally steep learning curve associated with L^AT_EX to ensure that anyone could create an outstanding thesis.

While the class file (`ualberta.cls`) deserves its own comprehensive documentation, this document will focus more specifically on the template file (`ualberta.tex`), as well as the following points:

- Installation and basic usage of L^AT_EX.
- Document structure and formatting.
- Inclusion of figures and tables.
- Inclusion of plots and graphs.
- Handling mathematical equations.
- Citations and references using BibLaTeX/BibTeX.
- Inclusion of Code and PDF's.
- Use of TeXstudio—L^AT_EX Editor.
- Use of JabRef—Reference Manager.
- Use of GitHub—Version Management.
- And more.

This guide does not cover advanced L^AT_EX programming or extensive customization of document classes. Instead, the class file `ualberta.cls` provides all the major document and formatting requirements as provided by GPS, while this document offers references on how to include the

various elements that might be required in a thesis. This includes all the explanations of the packages and macros needed to perform the examples.

1.4 Organization of the Guide

The guide is separated into multiple parts that contain a collection of chapters. The breakdown is as follows:

Part I: Writing a Thesis at the University of Alberta

- **Chapter 2:** University of Alberta Thesis Requirements
- **Chapter 3:** Submitting Your Thesis

Part II: Writing with uAlberta Thesis L^AT_EX Template

- **Chapter 4:** Getting Started with L^AT_EX
- **Chapter 5:** Document Structure and Formatting
- **Chapter 6:** Figures, Tables, & Plates
- **Chapter 7:** Plots, Charts, & Graphs
- **Chapter 8:** Mathematical Equations
- **Chapter 9:** Citations, References, and Cross-References

Part III: Technical Writing and Reference Management

- **Chapter 10:** TeXstudio: A Comprehensive IDE for L^AT_EX
- **Chapter 11:** JabRef: Managing Bibliographies Efficiently
- **Chapter 12:** Notepad++: A Lightweight Text Editor for Advanced Workflows

Part IV: Data Analysis and Programming

- a
- **Chapter 13:** GNU Octave: Open-Source Numerical Computation and Plotting

- **Chapter 14:** Git and GitHub: Robust Version Control for Your Thesis
- **Chapter 15:** Calcpad: Transparent Engineering Calculations

Part V: Images and Technical Figures

- a
- **Chapter 16:** ImageJ (Fiji): Scientific Image Processing
- **Chapter 17:** GIMP: Preparing Illustrative and Equipment Photographs
- **Chapter 18:** Dia: Structured Diagramming and Flowcharts

Each chapter provides detailed information, examples, and recommendations to help you navigate the thesis writing process within the $\text{\LaTeX}$ ecosystem.

1.5 Summary

This chapter introduced the background, objectives, scope, and organization of the guide. The subsequent chapters delve into specific topics, offering practical guidance and examples for mastering the art of writing a thesis in $\text{\LaTeX}$. Through this process, you will develop an understanding of how to create a thesis and manipulate content within the $\text{\LaTeX}$ ecosystem to produce an exceptional document.

Part I

Writing a Thesis at the University of Alberta

Chapter 2

University of Alberta Thesis Requirements

NOTICE I do recognize that there are other types of theses that are accepted at the University of Alberta, however, this document will only be focusing on the Chapter- and Paper-Based Theses.

The following Sections will review and comment on the current requirements based on the various documents provided by the Faculty of Graduate and Post-Doctorate Studies[0].

2.1 Title Page

The Faculty of Graduate and Post-Doctorate Studies provides effectively zero wiggle room on the title page, requiring it strictly follow the sample they provided [0].

2.2 Abstract

⚠ WARNING This is often considered one of the hardest sections to get right. This section is the summary of the entire work; you have just 700 words to convey your thesis, methodology, results, and conclusions while convincing the reader that the rest of the document is worth reading too. With all of that, the best piece of advice I can offer is to write this section last.

⚠ CAUTION Your abstract must be absent of figures, tables, references, and cross-references.

The abstract only has a few requirements:

- it must immediately follow the title page.
- it must be labelled as page “ii”.
- it must be double spaced.
- it must **not** include graphs, charts, tables, or illustrations.

2.3 Preface

⚠ WARNING The inclusion of a preface does not excuse a student from failing to acknowledge the contributions of others in the body of the thesis, as per the University’s Research and Scholarship Integrity Policy and the Code of Student Behaviour. One would still expect to see footnotes, endnotes, or in-text references within the thesis acknowledging the works.

⚠ CAUTION Acknowledgements, such as thanks to the supervisor and supervisory committee members, to colleagues, lab mates and friends, and to family, **do not** appear in the preface.

NOTICE If you need assistance on writing the preface, ask your supervisor. Your supervisor must review and verify the preface before it becomes part of the final version of the thesis.

The requirements for the preface [0]:

- Provide **full bibliographic citations** for any material in the thesis that has already been published and indicate the Chapter(s) where it appears.
- Acknowledge co-authorship or research collaboration that produced any of the material in the thesis and indicate the relative contributions of all collaborators and co-authors.

- Provide details of any ethics approval you obtained for the research.
- Acknowledge the use of generative Artificial Intelligence/Large Language Models if this has been permitted by your program and supervisory committee.
- Acknowledge third-party funding (grants, foreign governments, *etc.*) or,
- State that the thesis consists solely of your own unpublished and unfunded work.

The following sections will provide you with examples of how to include specific elements into your preface. The examples in the following sections have been modified from Faculty of Graduate & Postdoctoral Studies, *Appendix for thesis formatting guide*, version 4, University of Alberta, 2024. [Online]. Available: <https://www.ualberta.ca/en/graduate-studies/resources/appendix-for-thesis-formatting-guidelines-v4.pdf>. Due to the “Appendix for Thesis Formatting Guidelines” being generated using ChatGPT3.5 there are some inconsistencies within the document and the examples provided; while these are fine for providing a general idea of what to include, this goes against the ideology of this document. To combat this I have taken the liberty of adapting the examples to better blend into the style of this document and to increase the consistency, replacing some the generated examples with real examples where possible.

Full credit still goes to the University of Alberta’s Faculty of Graduate and Postdoctoral Studies for openly providing the document and I would strongly suggest anyone writing their thesis at the University of Alberta to read the updated documentation that they have put out in 2024. This new documentation provides much more guidance than the previous iteration which is no longer available.

Before diving right into the examples in [Sections 2.3.1 to 2.3.6](#), we will explore some of the still applicable guidelines for including your preface. First of which is a small change: since the FGSR became the GPS, the requirement of a preface has become **mandatory** regardless of the contents of the thesis.

When a thesis contains journal articles, the preface serves as a place for the student to include

a statement indicating his or her contribution to the journal articles. If any of the work presented in the thesis has led to any publications (accepted or published), these publications must be listed clearly in the preface with their bibliographical details and an indication as to where in the thesis this work is located (*e.g.* state in which Chapter or Chapters). For jointly authored publications¹, indication must also be given as to the relative contributions of the collaborators and co-authors, and a statement as to the proportion of research and writing conducted by the student. If ethics approval was required for the research, a statement to this effect must be included in the preface with the details of the approval that was granted.

Note: *Before preceding I would like to emphasize that the following examples are are not a one-and-done, you might need to combine elements from multiple of the following sections to build an appropriate preface for your thesis.*

i If **none** of the following apply **make sure** to include a preface in the style of the one shown in Section 2.3.6.

? Fun fact before you read on, Professor C. Ayranci who was mentioned in a few of the examples provided by *Appendix for Thesis Formatting Guide* was in-fact the supervisor for my Master's thesis.

2.3.1 Research Ethics Approval

If your thesis required you to get Research Ethics Approval, then you should include a preface based on the one here.

“This thesis is an original work by Daniel R. Aldrich. The research project, of which this thesis is a part of, received research ethics approval from the University of Alberta Research Ethics Board 3, Project Name ‘Etymologies and Entomologies: Unraveling the Threads of Language and Ecology’, No. 12345, February 23, 1993.”

¹Permission may be needed if the co-authors hold the copyright in these publications.

2.3.2 Collaborative Work

If your thesis required you to work collaboratively with other, organizations, researchers or otherwise, then you should include a preface based on the one here.

“Some of the research conducted for this thesis forms part of an international research collaboration, led by Professor T. Raivio at the University of Hogwarts, with Professor S. Agrawal being the lead collaborator at the University of Alberta. The technical apparatus referred to in [Chapter 5](#) was designed by myself, with the assistance of Professor A. Shiri and Professor C. Ayranci. The data analysis in [Chapter 6](#) and concluding analysis in [Chapter 7](#) are my own work, as well as the literature review in [Chapter 4](#).”

2.3.3 Previously Published Material

If your thesis is based on or includes work previously published by yourself, or work you co-authored, then you should include a preface based on the one here.

“[Chapter 4](#) of this thesis has been published as D. R. Aldrich *et al.*, ‘Osm-classic: An optical imaging technique for accurately determining strain,’ *SoftwareX*, vol. 6, pp. 225–230, 2017. DOI: [10.1016/J.SOFTX.2017.08.007](https://doi.org/10.1016/J.SOFTX.2017.08.007). I was responsible for the programming, data collection, and analysis as well as the manuscript composition. C. Ayranci assisted with the data collection by providing access to the Multi-Functional Composites Group’s Lab and Equipment, and contributed to manuscript edits. D. Nobes was the supervisory and corresponding author and was involved with concept formation and manuscript composition.”

2.3.4 Use of Artificial Intelligence (AI)

If your thesis used AI to help you outline, or otherwise write sections of your thesis; including help with analyzing, summarizing, *etc.*, then you should include a preface based on the one here.

“The generative artificial intelligence application or Large Language Model ChatGPT 3.5 was used for data analysis, summarization, synthesis, and simulation in [Chapter 5](#) of this thesis, as well as to generate a preliminary draft of the literature review in [Chapter 1](#).”

2.3.5 Receiving of Competitive Funding

If you received competitive funding used for your thesis, education, or research then you should include a preface based on the one here.

“This work was supported by a Doctoral Fellowship from the Social Sciences and Humanities Research Council, a grant from the Entomological Association of Edmonton, and the National Scholarship Council of Narnia.”

2.3.6 Previous Prefaces Did Not Apply

⚠ CAUTION

This was previously an *optional* inclusion in your thesis, however, since the FGSR transitioned to the GPS and have released new guidelines^[0] for formatting your thesis, this has now become **mandatory**.

If your thesis did not fall into any of the categories shown previously, then you should include a preface based on the one here.

“This thesis is an original work by YOUR FULL NAME. No part of this thesis has been previously published.”

2.4 Dedications or Quotes

NOTICE

Dedications or Quotations are limited to maximum of one page combined.

2.5 Acknowledgements

NOTICE

Acknowledgements are limited to maximum of two pages total.

The Acknowledgements is a strongly recommended, but not a strictly mandatory section of the thesis. For a thesis at the University of Alberta, an Acknowledgement page must be no more than two (2) pages in length.

The Acknowledgements page serves as a place within a thesis where students may wish to acknowledge the provision of funding from third parties, such as an external scholarship bodies, research granting agencies, and foreign governments. It is also appropriate to recognize the assistance provided by the supervisor and members of the supervisory committee. Additionally, you can recognize any help or support that you have received from family, friends, classmates, lab-mates, or any other individual that was able to help you in any way. A clear example of this would be the inclusion of the following, assuming that this template, class, or guide helped you in any significant way²:

“I would like to thank Daniel R. Aldrich for his continuing contributions to the University of Alberta, and for his work within the Graduate Student Community. More specifically, I would like to acknowledge the work that he has put into creating the L^AT_EX template that this thesis was created in and the ongoing support that he provides to the students at the University of Alberta in formatting their theses.”

2.6 Table of Contents

For your thesis at the University of Alberta, the Table of Contents must include all the headings including those of the optional components that are included in your thesis. This includes all the prefatory pages but excludes the title page. Additionally, you must include full chapter headings and 2–4 levels of subheadings.

²Please be aware that it is not necessary to include this statement in any way, shape, or form it is only provided as an example on how to include these type of statements; this template/guide is provided in good faith with the **only** expectation being that you **will** share it with your fellow colleagues, supervisor, and or anyone else that may benefit from this.

2.7 List of Figures, Tables, Plates, *etc.*

Include a list for each kind of non-textual item included in your thesis (figures, tables, plates, illustrations, videos, sounds files, *etc.*). Each list must start on its own page. Each list must be numbered sequentially, and be in a consistent format (*i.e.*, 1,2,3,... or 1.1,1.2,1.3,...).

2.8 List of Symbols or Abbreviations

Ask your supervisor or program director if your thesis needs a list of symbols or abbreviations. Each list must start on a separate page.

2.9 Glossary of Terms

Ask your supervisor or program director if this is necessary.

2.10 Other Preliminary Items

Any discipline- or thesis-specific prefatory materials not mentioned in these guidelines should be placed directly before the body of the thesis.

2.11 Body of the Thesis

Minimum requirement: introduction, presentation of your research, and conclusion. The first page is always numbered with Arabic numeral 1. All subsequent pages are numbered consecutively.

2.12 Bibliography, Works Cited, or References

The Bibliography³ has a few rules:

- it must be at the end of your thesis but before the appendices.
- it must **not** have a chapter number on it.

2.13 Appendices

Supporting material that is referenced in the body of the thesis. Appendices are not numbered as chapters, and should be kept to a minimum. Pages are numbered consecutively with the Bibliography.

References

- [0] Faculty of Graduate & Postdoctoral Studies, *Appendix for thesis formatting guide*, version 4, University of Alberta, 2024. [Online]. Available: <https://www.ualberta.ca/en/graduate-studies/resources/appendix-for-thesis-formatting-guidelines-v4.pdf>.
- [0] Faculty of Graduate & Postdoctoral Studies, *Gps minimum formatting requirements, Chapter- and paper-based theses*, University of ALberta, 2025. [Online]. Available: <https://docs.google.com/document/d/1WJmOdfgHFUN-q96q3ESCZnjXuX4VJS37MbD6bODPSoo/edit?usp=sharing>.
- [0] Faculty of Graduate & Postdoctoral Studies, *Thesis title page template*, University of Alberta, 2024. [Online]. Available: <https://www.ualberta.ca/en/graduate-studies/resources/2024-11-thesis-title-page-template.docx>.
- [0] Faculty of Graduate & Postdoctoral Studies, *Thesis submission checklist*, University of Alberta, 2025. [Online]. Available: <https://docs.google.com/document/d/1TmFbKVLbTPPJfP9aahXMvHPiIJY0pUzX4Fp1im1Iw88/edit?usp=sharing>.

³Note: you may choose to have references at the end of each chapter or included as part of a paper-based thesis. In these cases, there may be some duplication of the works cited depending on how they are included.

- [0] D. R. Aldrich, C. Ayranci, and D. S. Nobes, “Osm-classic: An optical imaging technique for accurately determining strain,” *SoftwareX*, vol. 6, pp. 225–230, 2017. DOI: [10.1016/J.SOFTX.2017.08.007](https://doi.org/10.1016/J.SOFTX.2017.08.007).

Chapter 3

Submitting Your Thesis

So you have seemingly gotten to the end of the writing, and you may be already taking the steps to set up your last review with your supervisor, set up your thesis defence, or even submit your thesis to GPS... but now what do you do?

Quick answer is: a lot; long answer: this will be discussed throughout this Chapter.

There are a number of steps that you will want to take to make sure that you are submitting the best version of your work. This includes checking for some of the more obvious and less obvious pitfalls that writing a Thesis in $\text{\LaTeX}$ or really any software poses.

3.1 Missing Cross-References and Citations

There are a couple of pitfalls when using $\text{\LaTeX}$ to automatically produce your cross-reference and citation. Because the compilation of the document is separated from the writing, it might not be obvious that `\cite{notakey}` or `\Cref{ch:notachapter}` are wrong until you compile the full document and check the document to see that they show up as **[notakey]** and **??**, respectively. However, it can happen that these become very easy to overlook in a document that can be well over 100+ pages.

We need a easy and straight forward way to make sure that all our cross-references and citations are correct. That is where our compiler’s log comes into play. Go to ‘Tools’ → ‘View Log’ to show the log under the editor in TeXstudio, then on the menu bar for the log un-select ‘show BadBox’. Now you should be left with a bunch of orange warnings; read through them and look for one where the message starts with “Reference” or “Citation” and ends in undefined. These are the problematic references or citations. You can click on the warnings to be taken to the file/line that the warning occurred on.

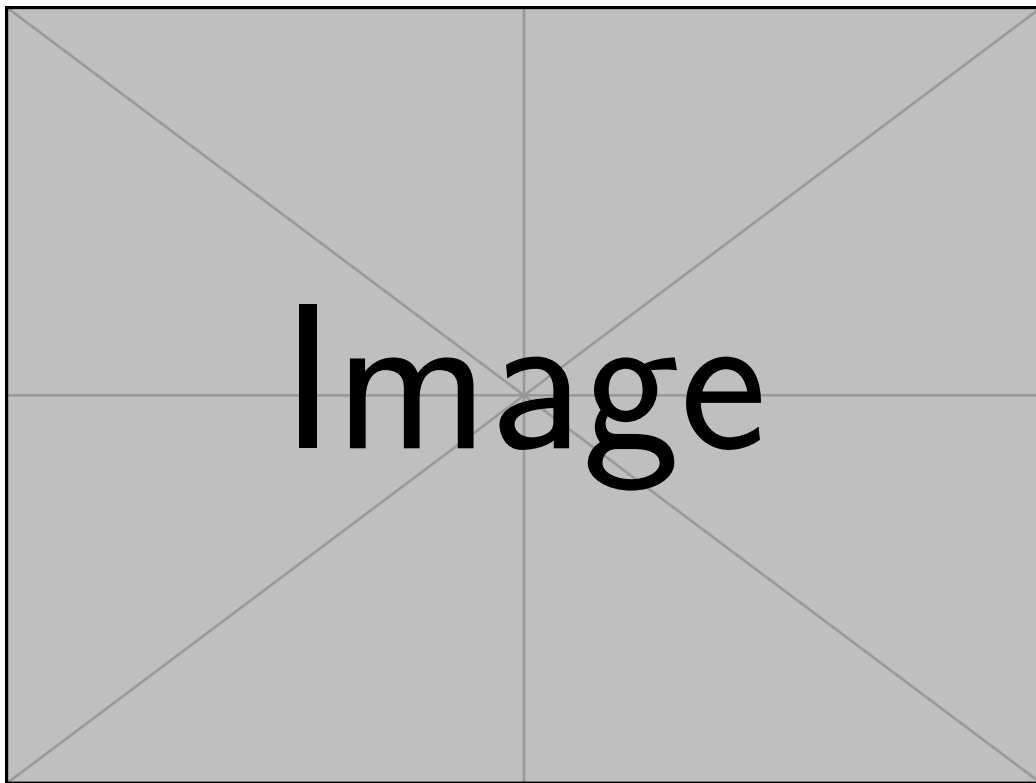


Figure 3.1: Example log in TeXstudio with badbox option turned off.

Part II

Writing with uAlberta Thesis L^AT_EX Template

The following part of the guide is focused on the practical application of L^AT_EX. Further, I will be showing you how to L^AT_EX within the scope of the provided template. Because, while learning L^AT_EX provides you with a powerful typesetting tool, the template acts as the “rulebook” that ensures your document will automatically adhere to the formatting guidelines set by the Faculty of Graduate and Post-Doctorate Studies (GPS).

Using the template with L^AT_EX effectively means you can spend your time on your research and data rather than tweaking margins, title page layout, or the ordering of front matter and other components of your thesis. These sections will walk you through the basic use of L^AT_EX and how the template, folder/file structure, and pre-defined macros/commands make the process of writing your thesis more streamlined.

I will be covering the workflow from the ground up:

- **Setup:** Installing L^AT_EX and the initialization of the template and understanding the “Main” document file (`ualberta.tex`).
- **Structure:** Organizing your chapters, abstracts, and inclusions, as well as utilizing the template’s commands for automatically generating the thesis’ front matter (Abstract, Preface, and Acknowledgements).
- **Content:** Best practices for inserting figures, tables, and complex equations.
- **Visuals:** Incorporating high-quality plots and graphs that maintain consistency in the final PDF.
- **Citations:** Managing a robust bibliography that integrates seamlessly with University of Alberta’s required citation requirements.

Chapter 4

Getting Started with L^AT_EX

4.1 Installation

To begin using L^AT_EX, you need to install a L^AT_EX distribution on your computer. Here there are two main distributors: [TeX Live](#), a free software distribution for the T_EX typesetting system which is provided by the T_EX User Group (TUG); or [MiKTeX](#), a free and open-source distribution of the T_EX/L^AT_EX typesetting system developed by Christian Schenk. Here are the links for installing L^AT_EX on different platforms:

4.1.1 Windows

For Windows users, you can install [TeX Live](#) or [MiKTeX](#). Download the installer from the respective websites and follow the installation instructions.

4.1.2 Mac

On Mac, you can use [MacTeX](#) (Mac version of TeX Live) or [MiKTeX](#). Download the package from the respective websites and follow the installation instructions.

4.1.3 Linux

For Linux users, [TeX Live](#) is a common choice or one can use [MiKTeX](#). Use your package manager to install it, or download the installer from the respective websites.

4.2 Basic Document Structure

Once L^AT_EX is installed, you can start by creating a basic L^AT_EX document. For a test of the system, open the included program **TeXworks**. Once it is open, copy the code below and press the typeset button (green play button). When the document has finished compiling you will get a preview of the document in a new window. Congratulations, you have compiled a new L^AT_EX document.

```
\documentclass{article}

\title{My First \LaTeX{} Document}
\author{Your Name}
\date{\today}

\begin{document}
  \maketitle

  Hello, \LaTeX{}!
\end{document}
```

This example demonstrates a simple L^AT_EX document with a title, author, and date. Now we will look a bit more into the example above and explain what exactly is happening with each line:

`\documentclass{article}` is a mandatory line that tells L^AT_EX the type of document to create; in this case, it is the built-in article class.

`\title{My First \LaTeX {} Document}` sets the title for the document to: My First L^AT_EX Document

`\author{}` sets the author of the document to: author.

`\date{\today }` sets the date for the document to: April 28, 2026

`\begin{document}` defines the beginning of the documents content.

`\maketitle` is a command that is defined in the article class (as well as other built-in classes) that generates the layout and adds the content from the `\title`, `\author`, and `\date` commands to the title of the document.

Hello, `\LaTeX{}`! generates the text: Hello L^AT_EX!

`\end{document}` defines the end of the documents content.

Now that we have installed L^AT_EX and have created our first document with it, we should explore a couple other pieces of software to streamline using L^AT_EX for creating a thesis. The next section will go over two main pieces of software typically used for technical writing in L^AT_EX: a L^AT_EX editor and a Reference Manager.

4.3 Other Software Considerations

While by itself L^AT_EX can be used with just a text editor and compiler (such as TeXworks), there are some additional software considerations that should be made. Mainly, the use of a dedicated **L^AT_EX editor** to help with the writing of the L^AT_EX “code” and the use of a **reference manager** to organize all your references.

4.3.1 L^AT_EX Specific Editor

While using a good text editor and compiling the document manually is all good, people new to L^AT_EX are not going to be too happy with it. So let's fix this by first getting you a sort of all-in-one program, a L^AT_EX IDE/editor.

While all of the distributions above come with [TeXWorks](#) (a simple editor that allows for

compilation) you might find it more beneficial to have some additional features. Most L^AT_EX editors have some form of editing, they help with the commands, they handle compilation, and viewing the final PDF. For a L^AT_EX editor I recommend one of the following:

- [TeXstudio](#)¹
- [Texmaker](#)
- [TeXnicCenter](#)
- [Overleaf](#)²

It is even possible to use popular coding editor [Visual Studio Code](#) with the [LaTeX Workshop](#) extension to edit and compile your document. Though I don't recommend the use of Visual Studio Code, as it is not feature complete like most of the editors above and may make some of the required steps harder to complete, or harder to diagnose any other issues that appear.

I am sure there are other editors besides the ones that I have listed, but any of these one will do you fine for writing your thesis.

4.3.2 Reference Manager

A reference manager is a database for streamlining the collection, management, and use of the references in your research. Instead of manually typing out every author name, journal title, and DOI into a `.bib` file, these programs allow you to save papers directly from your browser and organize them into folders.

For L^AT_EX users, the most important feature of a reference manager is the ability to export your library as a `.bib` file. This file contains all the metadata for your citations in a format that BibL^AT_EX or BibT_EX can understand. When you cite a paper in your document using its

¹This is the editor that I personally use, and will provide some tips on using the software in [Chapter 10](#).

²Due to changes with Overleaf, I recommend it less and less. While it is convenient, as it is a web based editor/compiler, they have restricted it more and more which forces the users to now pay more for the software... including to get longer compile times. While this is not an issue for short articles, in longer documents, such as a thesis, this can be a large hindrance and cost (~~\$100~~ \$109 CAD/year with student discount).

unique “citation key,” L^AT_EX looks into this file and formats the bibliography entry automatically according to your chosen style.

Here are a couple of the more popular reference managers available:

- [JabRef](#)³: An open-source manager designed specifically for BibT_EX. It uses the `.bib` format as its native file format, meaning there is no “exporting” required—you are working directly on the file L^AT_EX reads. Further, JabRef has support for creating a Microsoft Word reference library file.
- [Zotero](#): It is free, open-source, and has an incredible browser extension for saving papers with one click. For L^AT_EX users, the [Better BibT_EX](#) plugin is a must-have, as it automates the generation of citation keys and can keep your `.bib` file updated in real-time as you add new research.
- [Mendeley](#): A popular choice owned by Elsevier. While it has a polished interface and good PDF annotation features, it has become increasingly “closed off” in recent years, making it slightly more cumbersome to use with external L^AT_EX workflows compared to JabRef or Zotero.

Using one of these tools will save you dozens of hours during your thesis. Instead of worrying about where a comma goes in a citation, you can focus on the actual writing and the `.bib` file and L^AT_EX will handle all the proper formatting for you.

Note: *While a reference manager is suited for creating a thesis in L^AT_EX, it is not exclusive for use with L^AT_EX. In fact, even if you are writing your thesis in Microsoft Word or Google Docs I would recommend the use of the suggested reference managers to help keep track of all your references, and to keep track of the information you found within your references.*

³I personally use this as my reference manager, and will provide some tips on using the software in the [Chapter 11](#).

Chapter 5

Document Structure and Formatting

A thesis for the University of Alberta can consist of many different pieces that come together to create the final document. These will include the Title Page, Abstract, and other Prefatory pages; and the chapters, sections, and paragraphs; as well as the content, figures, tables, and other content you would like to include. In the rest of this chapter, we will look more specifically on the inclusion of the prefatory pages and how to structure your document with the various chapter, section, and paragraph commands with the help from the template.

5.1 Title Page, Abstract, and Other Prefatory Pages

As we saw in [Section 4.2](#), in $\text{\LaTeX}$ to create a title page you use the `\maketitle` command after providing the necessary title, author, and date information with the `\title{}`, `\author{}`, and `\date{}` commands, respectively. However, for a thesis at the University of Alberta a few more pieces of information are required for the title page. To help with this I have recreated the title page to match the sample provided by the Faculty of Graduate and Post-Doctorate Studies and have add the corresponding commands too. Additionally, a few more pages are also required (Abstract and Preface) and a few optional pages may be included too (Quote, Dedication, and

Acknowledgements). All of these pages have very specific formatting requirements provided by the GPS.

The Faculty of Graduate and Post-Doctorate Studies even provides a few different documents that attempt to provide the necessary information to create these pages [0]. While these do a decent job at layout how these pages need to be laid out, I decided to take the guess work completely out of it. I have created a few new pairs of commands; one sets the text for the section, the other one generates the page with the text in the proper formatting.

The following commands are used to **set the text** for each of their respective sections.

- `\abstracttext{}`,
- `\preface{}`,
- `\thesisquote{}`,
- `\dedication{}`, and
- `\acknowledgementtext{}`

Where the follow commands here are used to **print** the receptive sections.

- `\makeabstract`,
- `\makepreface`,
- `\makequote`,
- `\makededication`, and
- `\makeacknowledgements`

Because sometimes it can be easy to forget to comment or un-comment these, I have taken some time to make the system a bit smart. It will analyze if the commands that are used to set the text are empty or commented out and automatically exclude those sections or in the case of the preface it will default to the minimum required preface example, see [Section 2.3.6](#).

5.1.1 Title Page

The thesis Title Page has a few more fields to be filled in than a regular L^AT_EX document. These include `\degree`, `\specialization`, `\deptfac`, and `\convocationdate`. An example of how to fill these in can be seen in the original L^AT_EX code (`ualberta.tex`).

Most of the fields are fairly self-explanatory, however, to be extra clear as to what needs to be included I will now provide an explanation of each field:

`\title` this will be the title of your thesis.

`\author` this will be your full name.

`\degree` this is already set up to read from the file `selectDegree.tex` from the `00_LaTeX_Files` folder. The `selectDegree.tex` file is pre-setup with all the options to choose from, just un-comment the one that matches your thesis.

`\specialization` this will be where you write your specialization (if you have one) as it appears on beartracks.

`\deptfac` this is already set up to read from the file `selectDepartment.tex` from the `00_LaTeX_Files` folder. The `selectDepartment.tex` file is pre-setup with all the options to choose from, just un-comment the one that matches your thesis.

`\convocation` this is set up to read the current year, but if you will be convocating in the following year you must change this to match your convocation year.

`\creativecommons` this is already set up to read from the file `selectCreativeCommons.tex` from the `00_LaTeX_Files` folder. The `selectCreativeCommons.tex` file is pre-setup with all the options to choose from, just un-comment the one that matches your thesis.

NOTICE This class is not exclusive to the University of Alberta. To customize this to your specific institution, use the command `\university{YOUR INSTITUTION NAME}` before the `\maketitle` command. I have also made a lot of effort to comment and organize the `ualberta.cls` file to make editing as easy as possible.

5.1.2 Abstract

The template automatically includes and formats your abstract according to the Faculty of Graduate and Post-Doctorate Studies's requirements, using the `\makeabstract` command. To edit the abstract open the file `01_Prefatory/Abstract.tex` and add your text to the `\abstracttext` command.

5.1.3 Preface

The template automatically includes and formats your preface according to the Faculty of Graduate and Post-Doctorate Studies's requirements, using the `\makepreface` command. To edit the preface open the file `01_Prefatory/Preface.tex` and add your text to the `\preface` command. If it is left blank, with just a `{}` then the default mandatory preface will be used.

5.1.4 Dedication or Quotations

NOTICE Dedications or Quotations are limited to maximum of one page combined.

In the template provided, you may fill in the appropriate `Quote` and `Dedication` files from the prefatory folder: `01_Prefatory/`. If they are left blank, then they will automatically be excluded. Or you can just comment out the command `\setquotetext` or `\setdedicationtext` if you are unsure of whether you want to include one now or not and would just like to temporarily disable one or both of them. The command `\makededicationandquote` should never have to be commented out, as it will automatically detect the inclusion or exclusion of each and print the appropriate result.

5.1.5 Acknowledgements

In the template provided, you may fill in the appropriate Quote and Dedication fields in the document `01_Prefatory/Acknowledgements.tex`. If it is left blank, with just a `{}` then it will automatically be excluded.

The command `\makeacknowledgements` should never have to be commented out, as it will automatically detect the inclusion or exclusion of it and print the appropriate result.

5.1.6 Table of Contents

For your thesis, this template will automatically include all required chapters including the correct ones from the prefatory pages. However, due to stylistic considerations you may want more or fewer levels of subheadings included in the Table of Contents. To accomplish this I have created the command near the top of the `ualberta.tex` file that allows you to set both the depth of the Numbered Headings and the level of depth for the Table of Contents. To do this change the number, `n`, in the `\settoclevel{n}` command to 2, 3, or 4 for two, three, or four levels of subheading, respectively.

💡 For the `\settoclevel{n}` command, setting `n` to 2 will include everything down to subsection, 3 will include everything down to subsubsection, and 4 will include everything down to paragraph in the Table of Contents.

5.1.7 Lists of Figures, Tables, ...

This template is setup to for figures, tables, plates, illustrations, photos, videos, and sound files. Each of items has their own environment `figure`, `table`, `plate`, `illustration`, `photo`, `video`, and `sound`, respectively. These following set of smart-commands will automatically include or exclude the respective “List of ...” in the prefatory pages, no need to comment or uncomment them. If you would like them to appear in a different order, just change their order in the the

ualberta.tex file.

- `\listoffigures`
- `\listoftables`
- `\listofplates`
- `\listofillustrations`
- `\listofvideos`
- `\listofphotos`
- `\listofsounds`

Note: *None of these commands need to be commented out, they can stay un-commented, and as long as $\LaTeX$ is run at minimum twice (as is required for BibLaTeX/BibTeX, and any reference updates) they will update themselves to be included or excluded automatically.*

5.2 Nomenclature, Glossary, & Acronyms

5.2.1 Lists of Symbols

The `\listofsymbols` command will print the List of Symbols.

```
\nomenclature[CATEGORY]{SYMBOL}{DEFINITION}
```

Adding the `\nomunit` command inside and at the end of the **DEFINITION** will add its **CONTENTS** to the end of the line with the content being right justified. `\nomunit{CONTENTS}`

To add more categories to sort the Nomenclature or List of Symbols by you just need to add a new line to the code, show in **Listing 5.1**, at the top of the `Nomenclature.tex` file in `01_Prefatory` folder.

Listing 5.1: Nomenclature Grouping Settings

```
1 \renewcommand\nomgroup[1]{%
```

```

2   \item[\bfseries%
3       \ifstrequal{#1}{A}{Constants}{}%
4       \ifstrequal{#1}{B}{Latin}{}%
5       \ifstrequal{#1}{C}{Greek}{}%
6       \ifstrequal{#1}{D}{Subscripts}{}]%
7

```

5.2.2 Lists of Abbreviations

The `\abbreviations` command will print the List of Symbols.

```
\addacronym{ACRONYM}{DEFINITION}
```

5.2.3 Glossary of Terms

The `\generateglossary` command will print the List of Symbols.

```
\addterm{TERM}{DEFINITION}
```

5.3 Chapters, Sections, Subsections, *etc.*

Organize your document hierarchically using chapters, sections, subsections, *etc.* These structures all utilize the base macros from L^AT_EX including:

- `\chapter{Chapter Heading}`,
- `\section{Section Heading}`,
- `\subsection{Subsection Heading}`,
- `\subsubsection{Sub-Subsection Heading}`,
- `\paragraph{Paragraph Heading}`, and
- `\subparagraph{Subparagraph Heading}`.

- 💡 For writing your thesis, it is strongly recommended that one outlines the thesis using these commands first, while also added in a small description of what that chapter, section, *etc.*, should accomplish. This will help you stay organized and on track. Remembering that you can use comments, `%`, to hide these descriptions when you start to fill in your content.

5.4 Page Layout and Margins

⚠ WARNING

While one can adjust the values using the commands provided by the following packages, unless you really know $\text{\LaTeX}$ inside and out this should be avoided. Everything how I have provided in these files is aimed at making writing your thesis as easy as possible including the options I have chosen for the included packages.

You can customize the layout and margins of your document using the `geometry` package. Additionally, you can use the `titlesec` package to customize the formatting of chapter and section titles.

References

- [0] Faculty of Graduate & Postdoctoral Studies, *Appendix for thesis formatting guide*, version 4, University of Alberta, 2024. [Online]. Available: <https://www.ualberta.ca/en/graduate-studies/resources/appendix-for-thesis-formatting-guidelines-v4.pdf>.

Chapter 6

Figures, Tables, & Plates

WARNING

Throughout this Chapter there will be code listings that include lines that show the required packages. These lines should be included in your $\text{\LaTeX}$ preamble, not in the body of your document. To make this easy I have actually included a document specifically to add your packages too. This can be found in the `includePackages.tex` file found in the `00_LaTeX_Files` folder.

 `./00_LaTeX_Files/includePackages.tex` includes all the packages required for creating all the examples in this document, some of these will not be necessary for your own thesis, however, I have included comments on all packages for what they are used for. This allows you to make a decision on if you will need them as well. If you are unsure, you may just comment out the line so as to not forget the packages that were originally included.

6.1 Introduction

Figures, Tables, and Plates play a crucial role in conveying complex information effectively within academic documents. This chapter will delve into the intricacies of incorporating these visual elements into your $\text{\LaTeX}$ document, exploring both basic features and advanced techniques to enhance the visual appeal and clarity of your content. While there are many types of figures that one might include in a document, this Chapter specifically focuses on the inclusion of graphic

figures (pictures) and the layout of multi-figures (sub-figures).

6.2 Inserting Figures

In L^AT_EX, figures are included using the `graphicx`¹ package. The `\includegraphics` command is used to insert an image. Let's consider an example:

```
1 \begin{figure}[htbp]
2   \centering
3   \includegraphics[width=0.45\linewidth]{example-image}
4   \caption{Example Figure}
5   \label{fig:example}
6 \end{figure}
```

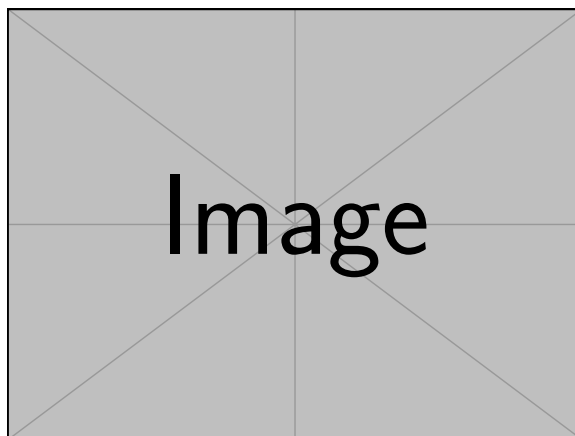


Figure 0: Example Figure

In this example, the `figure` environment is used to contain the image. The optional argument `[htbp]` for the `figure` environment controls the order in which L^AT_EX will attempt to insert the float (figure).

- `h` tries to place the float **here**.
- `t` tries to place the float at the **top** of the page.

¹D. Carlisle *et al.*, *Packages in the ‘graphics’ bundle*, version 1.2e, 2025. [Online]. Available: <https://ctan.org/pkg/graphicx>.

- `b` tries to place the float at the **bottom** of the page.
- `p` tries to place the float on a float only **page**.

The `\centering` command ensures the image is centred horizontally. The `width` parameter is used to control the size of the image; in this case, it has been set to `0.45\linewidth` which will make it fill a space that is 0.45 times (45%) the width of the current line. The `\caption` and `\label` commands provide a caption and label for referencing, respectively.

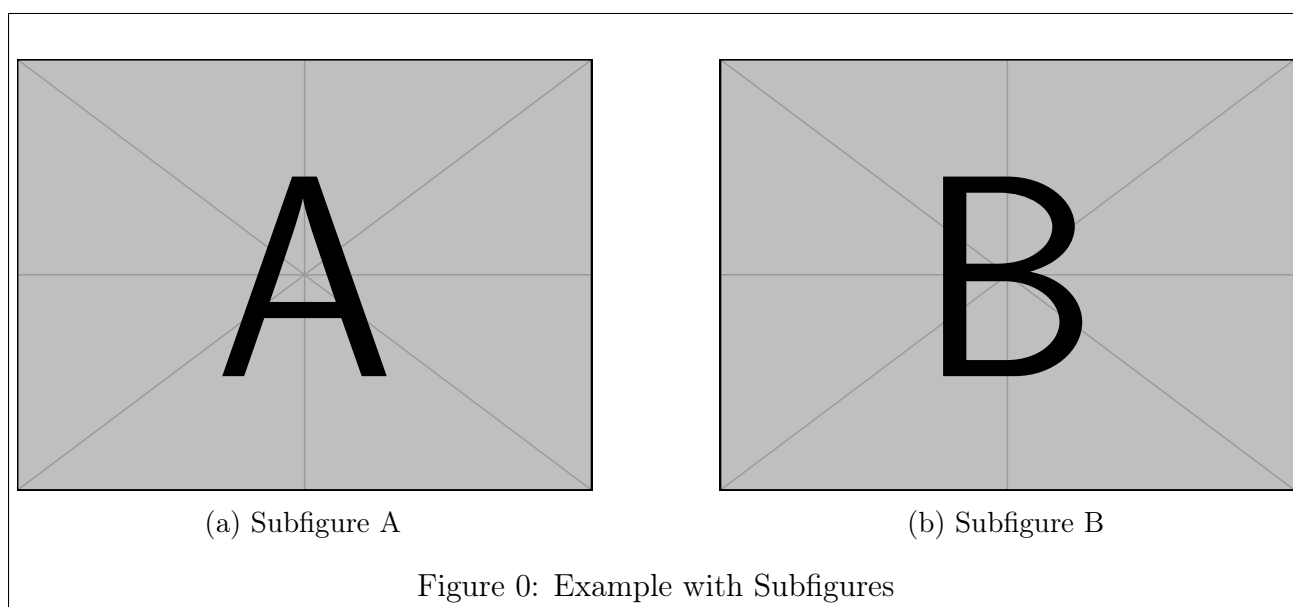
Figures can be formatted to meet specific requirements. The `\subfigure` command from the `subcaption`² package can be used for side-by-side figures:

²A. Sommerfeldt, *The subcaption package*, version 1.6, 2023. [Online]. Available: <https://ctan.org/pkg/subcaption>.

```

1 \begin{figure}[htbp]
2   \begin{subfigure}{0.45\linewidth}
3     \centering
4     \includegraphics[width=\linewidth]{example-image-a}
5     \caption{Subfigure A} % Leave blank for just letter
6     \label{subfig:a}
7   \end{subfigure}
8   \hfill
9   \begin{subfigure}{0.45\linewidth}
10    \centering
11    \includegraphics[width=\linewidth]{example-image-b}
12    \caption{Subfigure B} % Leave blank for just letter
13    \label{subfig:b}
14  \end{subfigure}
15  \caption{Example with Subfigures}
16  \label{fig:subfigures}
17 \end{figure}

```



This example uses the `subfigure` environment to create subfigures within a larger `figure` environment. The `\hfill` command adds horizontal space between the subfigures.

While this section provided a few examples on how to make some figures and subfigures, I would strongly recommend checking out some of the more complex examples of figures shown in [Appendix A](#). That appendix provides the exact code and examples for creating much more intricate subfigure grids and layouts.

6.3 Tables and Tabularx

Tables in $\text{\LaTeX}$ are traditionally created using the `tabular` environment. However, the `tabularx`³ package is particularly useful when you want the table to automatically calculate and adjust its column widths to fit a specific dimension. To make writing tables a little easier, let's define some custom column types in our `includeMacros.tex` file:

```
1 \usepackage{tabularx}
2 \newcolumntype{C}{>{\centering\arraybackslash}X}
3 \newcolumntype{L}{>{\raggedright\arraybackslash}X}
4 \newcolumntype{R}{>{\raggedleft\arraybackslash}X}
```

Now, let's implement a table utilizing the `tabularx` custom columns:

```
1 \begin{table}[htb]
2   \centering
3   \begin{tabularx}{\linewidth}{|C|L|R|}
4     \hline
5     \textbf{Centered} & \textbf{Left Justified} & \textbf{Right Justified} \\
6     \hline
7     Content & More content & Additional content \\
8     \hline
9   \end{tabularx}
10  \caption{Example Table with Tabularx}
11  \label{tab:example}
12 \end{table}
```

Centered	Left Justified	Right Justified
Content	More content	Additional content

Table 0: Example Table with Tabularx

In this example, the `tabularx` environment is used, and the custom column types `C`, `L`, and `R` are applied to the columns. This ensures the content is cleanly centered, left-justified, and right-justified, respectively

³D. Carlisle, *The tabularx package*, version 2.12a, 2023. [Online]. Available: <https://ctan.org/pkg/tabularx>.

6.4 Advanced Table Features

To create professional-looking, publication-ready tables, the `booktabs`⁴ package is the industry standard. It provides specialized commands for better vertical spacing and styling of table rules (lines):

```

1 \begin{table}[htbp]
2   \centering
3   \begin{tabular}{ccc}
4     \toprule
5     \textbf{Header 1} & \textbf{Header 2} & \textbf{Header 3} \\
6     \midrule
7     Content 1 & Content 2 & Content 3 \\
8     Content 4 & Content 5 & Content 6 \\
9     \bottomrule
10  \end{tabular}
11  \caption{Example Table with Booktabs}
12  \label{tab:booktabs_example}
13 \end{table}

```

Header 1	Header 2	Header 3
Content 1	Content 2	Content 3
Content 4	Content 5	Content 6

Table 0: Example Table with Booktabs

By replacing standard `\hline` commands with `\toprule`, `\midrule`, and `\bottomrule`, you create horizontal rules with appropriate, aesthetically pleasing spacing that adheres to formal typesetting guidelines.

⁴S. Fear, *Publication quality tables in latex*, version 1.61803398, 2020. [Online]. Available: <https://ctan.org/pkg/booktabs>.

6.5 Additional Packages for Enhanced Table Functionality

Depending on your data, several other packages can be employed to heavily enhance table functionality:

- The `longtable`⁵ package allows tables to seamlessly span across multiple pages, which is essential for displaying large datasets.
- The `multirow`⁶ package provide targeted commands for merging cells that need to span across multiple rows.
- The `makecell`⁷ package enables much more complex table cell layouts, such as forced line breaks within a single cell.

Each of these packages comes with its own set of commands and options. Let's briefly explore the usage of `longtable`, `multirow`, and `makecell`:

In these examples, the `longtable` environment manages tables that span multiple pages by defining headers and footers that repeat. The `\multirow` command is employed to create cells that span vertically across multiple rows, while `\multicolumn` is used for cells that span horizontally across multiple columns.

6.6 Conclusion

This Chapter provided a comprehensive overview of including figures, plates, and tables in your L^AT_EX document. From the basic insertion of figures to advanced table formatting using powerful packages like `tabularx`, `booktabs`, and others, you now have a solid foundation to manage visual data in your thesis. While this chapter provided a lot of details on how to add figures that

⁵D. Carlisle, *The longtable package*, version 4.24, 2025. [Online]. Available: <https://ctan.org/pkg/longtable>.

⁶P. van Oostrum, *The multirow, bigstrut and bigdelim packages*, version 2.9, 2024. [Online]. Available: <https://ctan.org/pkg/multirow>.

⁷O. Lapko, *The makecell package*, version 0.1e, 2009. [Online]. Available: <https://ctan.org/pkg/makecell>.

have already been pre-generated, you might want to generate figures on the spot-potentially using data generated directly from other programs (such as MatLab[®], Python, *etc.*). For this advanced topic, [Chapter 7](#) provides the in-depth workings of the `pgfplots` package and teaches you how to generate consistent, professional-looking plots and graphs programmatically.

References

- [0] D. Carlisle, S. Rahtz, and The LaTeX Project Team, *Packages in the ‘graphics’ bundle*, version 1.2e, 2025. [Online]. Available: <https://ctan.org/pkg/graphicx>.
- [0] A. Sommerfeldt, *The subcaption package*, version 1.6, 2023. [Online]. Available: <https://ctan.org/pkg/subcaption>.
- [0] D. Carlisle, *The tabularx package*, version 2.12a, 2023. [Online]. Available: <https://ctan.org/pkg/tabularx>.
- [0] S. Fear, *Publication quality tables in latex*, version 1.61803398, 2020. [Online]. Available: <https://ctan.org/pkg/booktabs>.
- [0] D. Carlisle, *The longtable package*, version 4.24, 2025. [Online]. Available: <https://ctan.org/pkg/longtable>.
- [0] P. van Oostrum, *The multirow, bigstrut and bigdelim packages*, version 2.9, 2024. [Online]. Available: <https://ctan.org/pkg/multirow>.
- [0] O. Lapko, *The makecell package*, version 0.1e, 2009. [Online]. Available: <https://ctan.org/pkg/makecell>.

```

\usepackage{longtable}
\usepackage{multirow}
\usepackage{multicolumn}

% Example Longtable
\begin{longtable}{|c|c|}
  \caption{Longtable Example} \label{tab:longtable} \\
  \hline
  \textbf{Header 1} & \textbf{Header 2} \\
  \hline
  \endfirsthead
  \hline
  \textbf{Header 1} & \textbf{Header 2} \\
  \hline
  \endhead
  Content 1 & Content 2 \\
  Content 3 & Content 4 \\
  \hline
\end{longtable}

% Example Multirow and Multicolumn
\begin{table}[htb]
  \centering
  \begin{tabular}{|c|c|c|}
    \hline
    \multirow{2}{*}{\textbf{Multirow-Col1}} & \multicolumn{2}{c|}{\textbf{Multicolumn-Col2-3}} \\
    \cline{2-3}
    & \textbf{Column 2} & \textbf{Column 3} \\
    \hline
    Content 1 & Content 2 & Content 3 \\
    \hline
  \end{tabular}
  \caption{Example Table with Multirow and Multicolumn}
  \label{tab:multirow_multicolumn}
\end{table}

```

Chapter 7

Plots, Charts, & Graphs

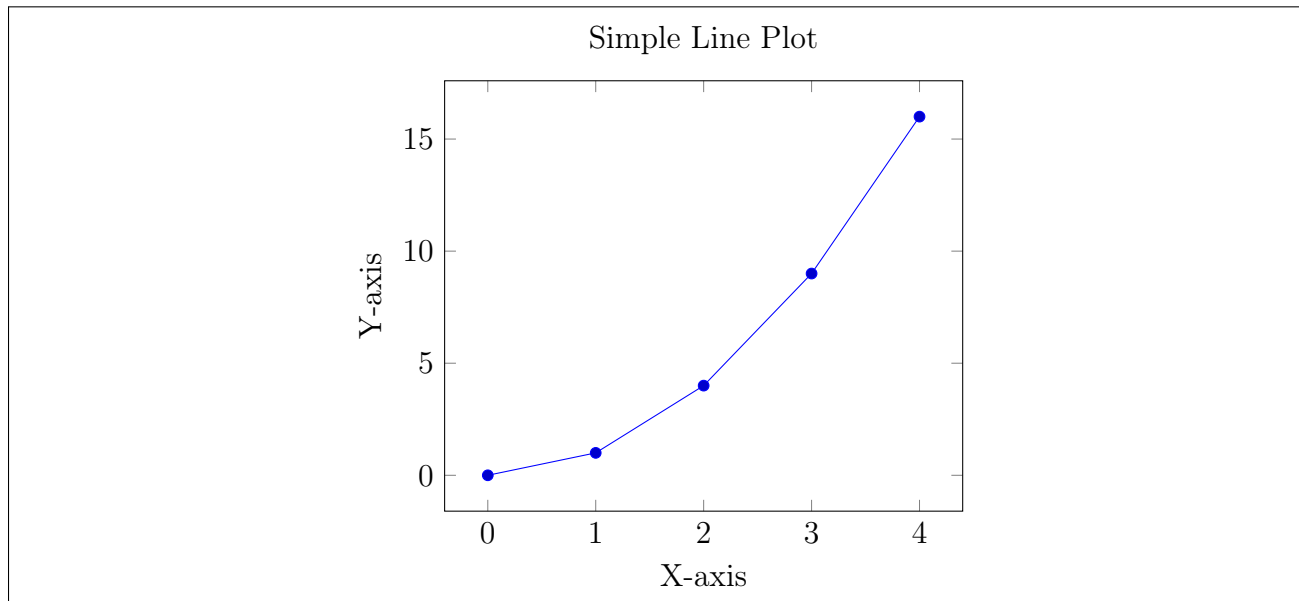
Throughout this chapter we will be exploring some of the different ways of displaying your data in your thesis. Mainly this will be accomplished with the `pgfplots`¹ package. In the following sections, there will be a few examples of how to generate different plots. For more information on how to create plots, [here](#) is the manual for `pgfplots` (the package used to generate the information for TikZ to create the plots). For an extensive list of examples please refer to [this](#)

7.1 Line Plots

A simple line plot can be effectively created using the `axis` environment from the `pgfplots` package in $\text{\LaTeX}$. The `pgfplots` package is a powerful tool for creating high-quality plots directly within $\text{\LaTeX}$ documents. It builds upon the `TikZ` package and provides a comprehensive set of options for plotting and customizing graphs.

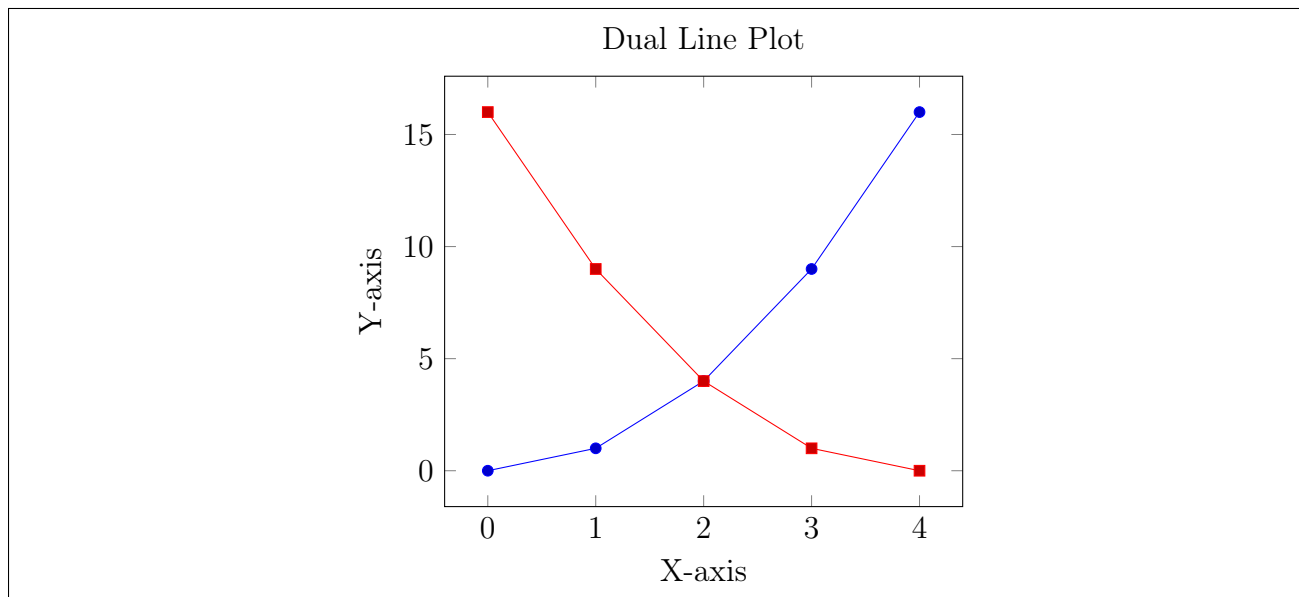
Expanding on this example, we can add a second plot by adding the code shown below [Figure 7.2](#). This will result in the addition of the second red line.

¹`pgfplots`.



```
1 \begin{tikzpicture}
2   \begin{axis}[
3     title={Simple Line Plot},
4     xlabel={X-axis},
5     ylabel={Y-axis},
6   ]
7     \addplot coordinates {
8       (0,0)
9       (1,1)
10      (2,4)
11      (3,9)
12      (4,16)
13    };
14   \end{axis}
15 \end{tikzpicture}
```

Figure 7.1: A simple line plot and the code to generate the plot



```

1 \begin{tikzpicture}
2   \begin{axis}[
3     title={Dual Line Plot},
4     xlabel={X-axis},
5     ylabel={Y-axis},
6   ]
7     \addplot coordinates {
8       (0,0)
9       (1,1)
10      (2,4)
11      (3,9)
12      (4,16)
13    };
14    \addplot coordinates {
15      (0,16)
16      (1,9)
17      (2,4)
18      (3,1)
19      (4,0)
20    };
21  \end{axis}
22 \end{tikzpicture}

```

Figure 7.2: A simple line plot with two sets of data.

7.2 Customizing Plots

This section provides some ways to increase the readability and customization of the plots we generate.

7.2.1 Increasing the Plot Size

7.2.2 Adding a Legend

Legends can be added to plots for better readability. To add a legend to your plot you can use the code in [Section 7.2.2](#) to generate the plot shown in [Figure 7.3](#). Each plot is individually added to the legend by adding a `\addlegendentry{YOUR LEGEND ENTRY HERE}` command following the `\addplot` command.

Note: *The position of the legend can be specified by using the optional parameter `legend pos=` followed by a set of compass coordinates.*

```
\begin{figure}[H]
  \centering
  \begin{tikzpicture}
    \begin{axis}[
      title={Plot with Added Legend},
      xlabel={X-axis},
      ylabel={Y-axis},
      legend pos=north west,
    ]
```

```
\addplot coordinates {
    (0,0)
    (1,1)
    (2,4)
    (3,9)
    (4,16)
};
\addlegendentry{\(y = x^2\)}
\addplot coordinates {
    (0,16)
    (1,9)
    (2,4)
    (3,1)
    (4,0)
};
\addlegendentry{\(y = 16 - x^2\)}
\end{axis}
\end{tikzpicture}
\caption{A customized plot with a legend.}
\label{fig:legend-plot}
\end{figure}
```

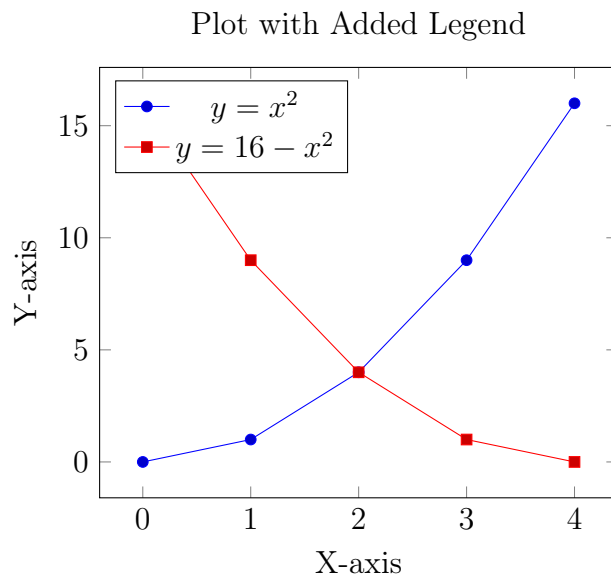


Figure 7.3: A customized plot with a legend.

7.2.3 Adding Grid Lines

To add gridlines to your plot

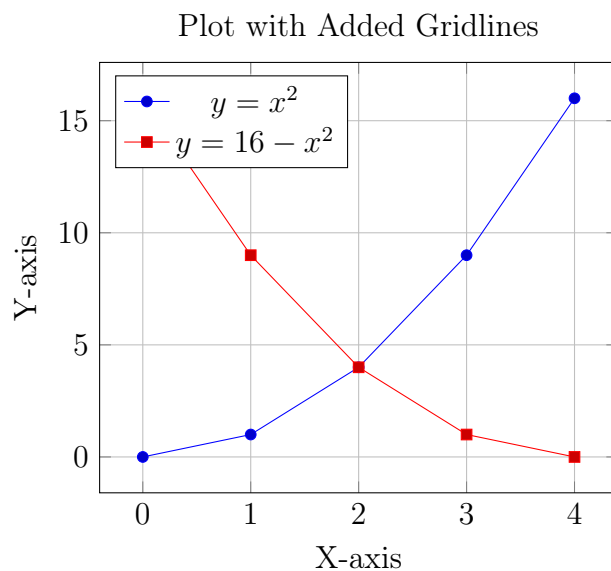


Figure 7.4: A customized plot with added gridlines.

7.2.4 Changing Colors and Line Styles

Colors and line styles can be easily modified:

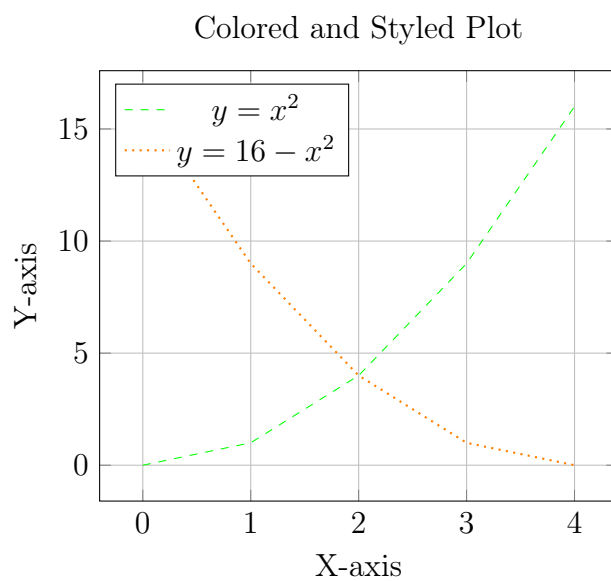


Figure 7.5: A plot with customized colors and line styles

7.3 Advanced Plot Types

7.3.1 Equations

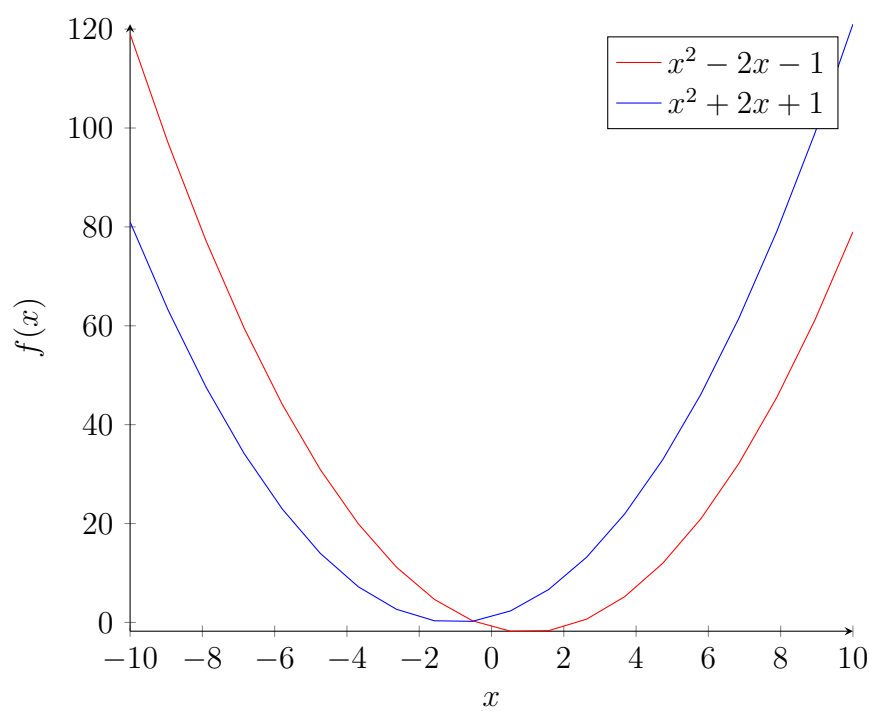


Figure 7.6: Plot of two parabola.

7.3.2 Scatter Plot with External Data

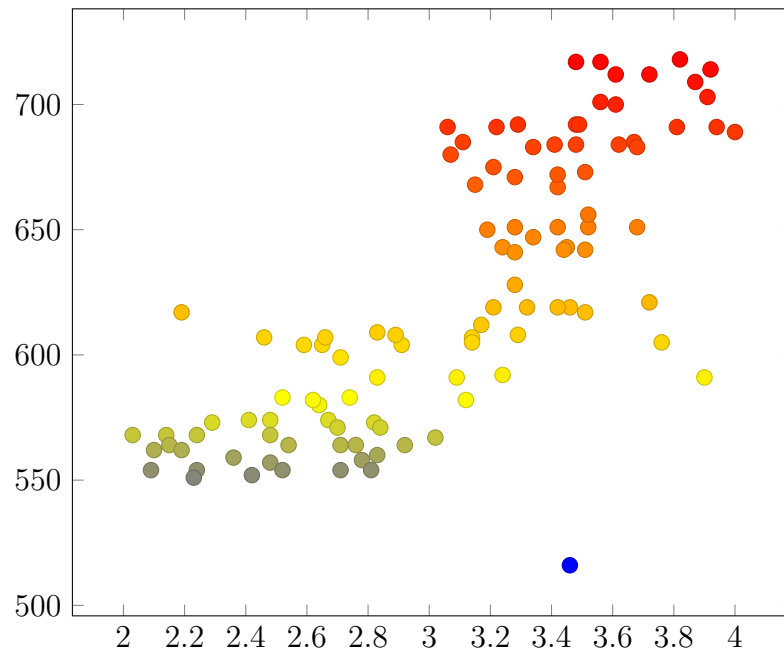


Figure 7.7: Example of a Scatter Plot.

7.3.3 Bar Plot

Bar plots are useful for categorical data. Here's how to create one:

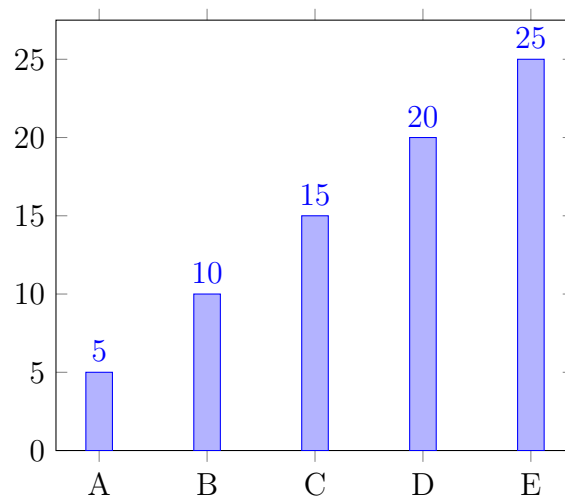


Figure 7.8: A bar plot

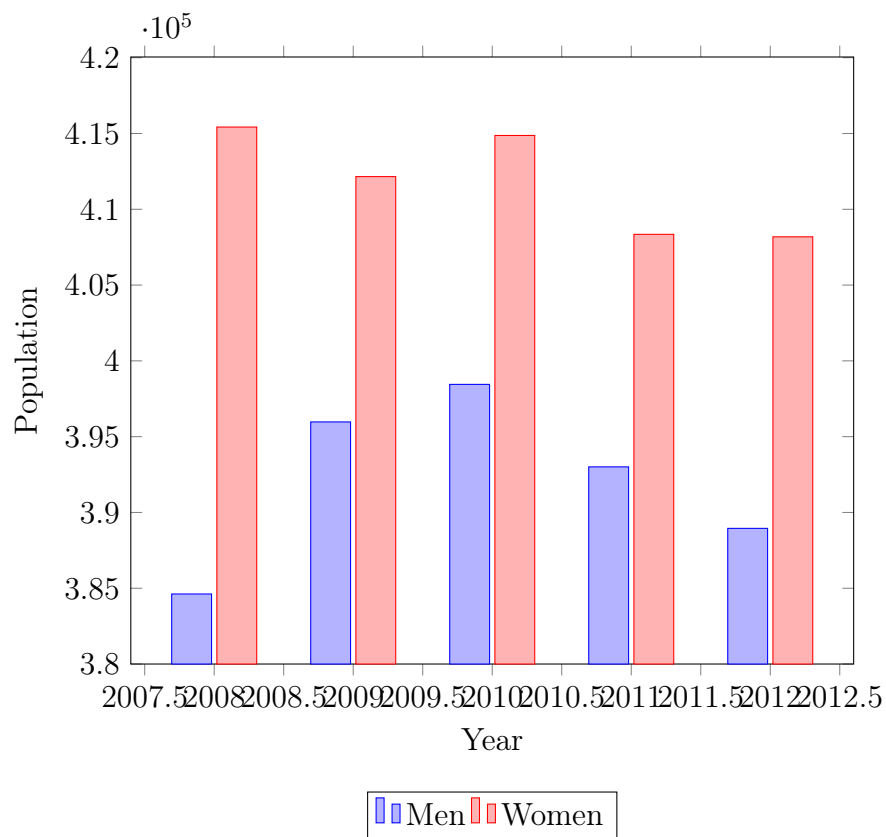


Figure 7.9: Example of a Bar Graph.

7.3.4 Pie Chart

Pie charts are less common in $\text{\LaTeX}$, as well as, a lot of other media due to the poor representation of the data. However, if you are still inclined to use them the remainder of this section will show how the package `pgf-pie` can be used to create consistent graphs that look good.

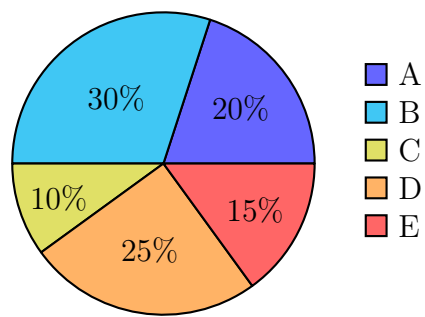


Figure 7.10: A basic pie chart.

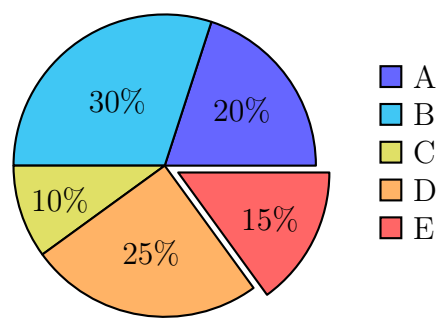


Figure 7.11: A pie chart with an “Exploded” slice.

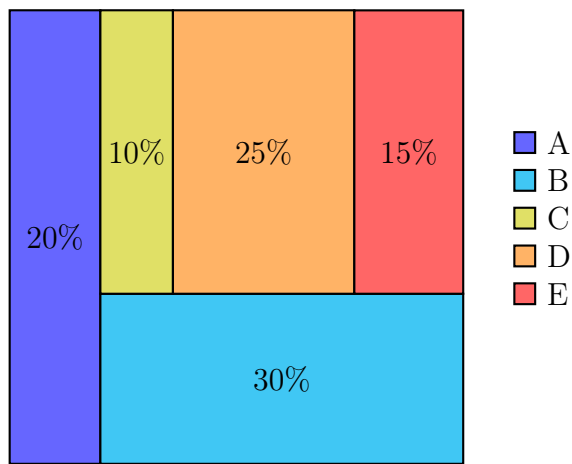


Figure 7.12: A “square” pie chart.

7.3.5 3D Plot

3D plots can be created for more complex data visualization:

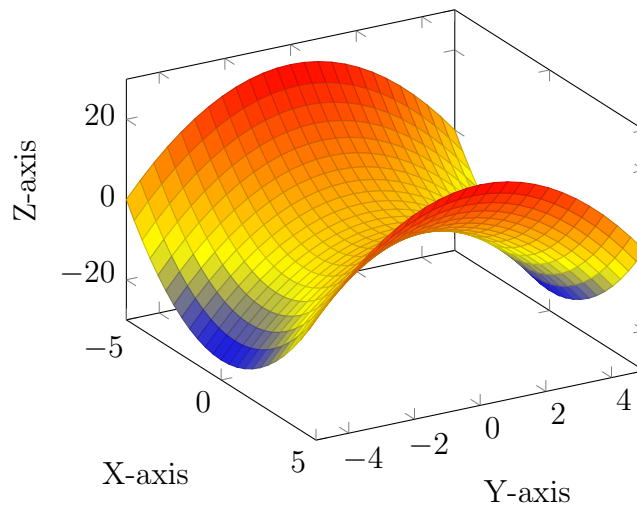


Figure 7.13: A 3D surface plot

Example using the mesh parameter

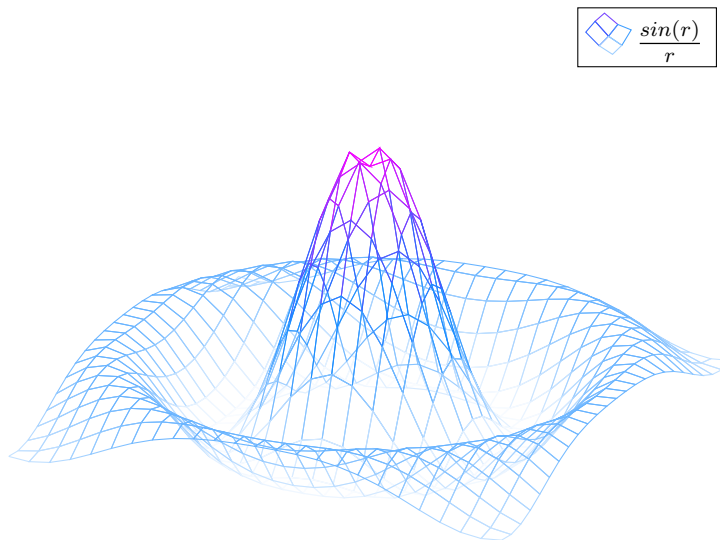


Figure 7.14: Example of a 3D Plot

7.3.6 Polar Plot

Polar plots are useful for circular data:

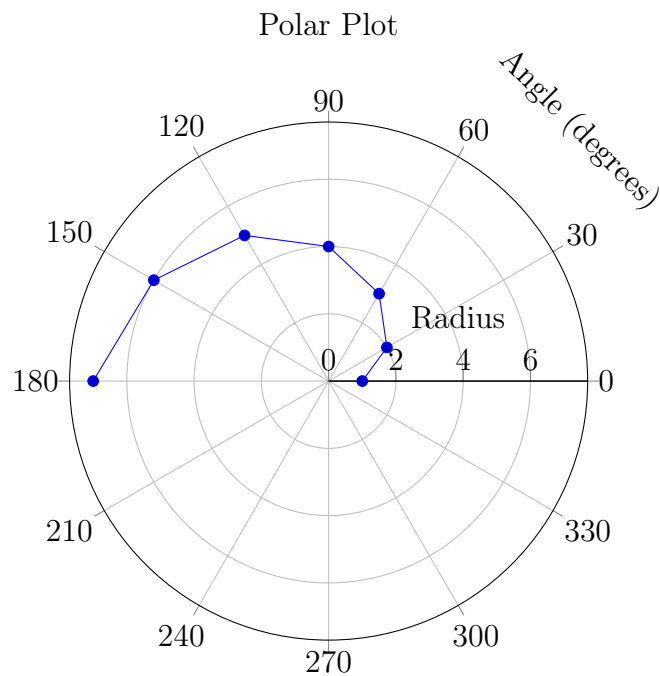


Figure 7.15: A polar plot

7.3.7 Box Plot

Box plots are used to visualize the distribution of data:

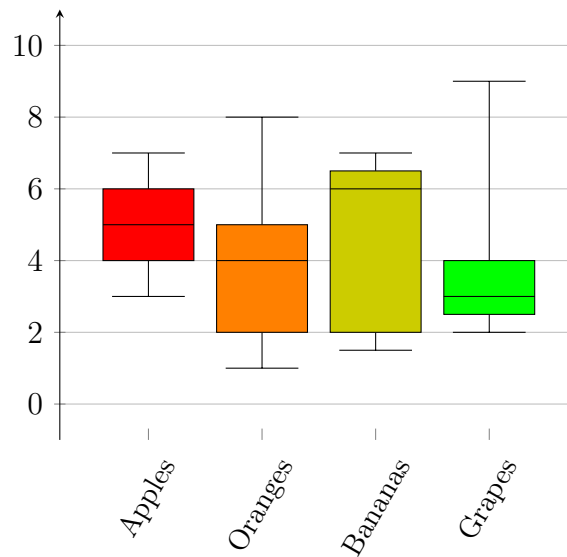


Figure 7.16: A box plot

7.4 Conclusion

The `pgfplots` package is an incredibly versatile tool for creating a wide range of plots and graphs in $\text{\LaTeX}$. This chapter has provided examples of various plot types and customization options, showcasing the power and flexibility of `pgfplots`. By leveraging these capabilities, you can create high-quality, publication-ready figures for your thesis.

Chapter 8

Mathematical Equations

There are many ways to include formulas in your thesis. This section will provide some different ways of adding them (inline and standalone), as well as provide some ways of referencing the equations.

To start the simplest way to add an equation is using the built-in L^AT_EX math mode. To enter and exit math mode one just needs to use the `\(` and `\)` symbols around an equation. While there also exists `$<Equation>$` to add math, it is not recommended due to potential compatibility issues. Additionally, this, `\(<Equation>\)`, method is capable of being redefined to add further customization. An example of using math mode to get an inline equation is by using the following command:

```
\(\vec{F_{d}}=\frac{1}{2} A C_d \vec{V}^2\)
```

The above command has the effect of creating the following output: $\vec{F_d} = \frac{1}{2} A C_d \vec{V}^2$. Sometimes it can be quite beneficial to separate what would be an inline equation to be on its own line. For this, we have two different ways of doing it. The first way will produce an equation that has no reference:

$$E = m c^2$$

$$E = m c^2$$

The second will produce an equation with a reference. For this, there are two main ways of creating the reference, the first one, see [Equation \(8.1\)](#), creates a numbered reference; the other one, see [Equation \(Constant pi\)](#), creates a reference with a ‘tag’. The difference between the two is the inclusion of a `\tag{<text>}` command that will replace the regular number with `<text>` and the changing from the `equation` environment to the `equation*` environment. If you do not want the brackets around the tag, as shown in [Equation \(Constant pi\)](#), use the starred version of the command: `\tag*{<text>}`. This will not remove the braces in the reference for the equation, but will remove them from appearing next to the equation definition.

NOTICE Using the `\tag` command in conjunction with the `\label` command can create a L^AT_EX warning when used in the non-star `equation` environment. This warning can be safely ignored, however, the better way to deal with this is to make sure one is using the star version, `equation*`.

$$\pi = 3.14\dots \tag{8.1}$$

$$\pi = 3.142\dots \tag{Constant pi}$$

$$\pi = 3.1415\dots \tag{Constant pi}$$

If you have multiple equations that you want arranged very neatly, use the `align` environment and you can assign individual equations numbers as shown in [Equations \(8.2\) to \(8.4\)](#). Note that it is the `&` symbol that determines what will be aligned. Further note that spaces in “math

mode” are ignored and need to be specified using the space commands in

$$\textit{Equation1} = 1 \tag{8.2}$$

$$\textit{Equation2} = 2 + 2 \tag{8.3}$$

$$\textit{Equation3} = 3 + 3 + 3 \tag{8.4}$$

It may be very important in a math heavy thesis to be able to show your equations, or even data in a readable way. For this, we will explore some of the ways to create specific data.

8.1 Vector, Sets, Piecewise Functions, Matrix Math, and More

$$f(x) = \begin{cases} x^{2*\ln x}, & \text{if } x < 3 \\ -\frac{x}{2}, & \text{if } 3 \leq x \leq 4 \\ x, & \text{if } 4 < x \end{cases} \tag{8.5}$$

Vectors and Matrices are used in many fields of math and science and provide a convenient

way to represent 2-Dimensional arrays of numbers.

$$x \in \{1, 2, 3, 4, 5, 6, 7\} \tag{8.6}$$

$$V_1 = \left(\begin{array}{cccc} a, & b, & c, & d \end{array} \right) \tag{8.7}$$

$$V_2 = \left(\begin{array}{c} a \\ b \\ c \\ d \end{array} \right) \tag{8.8}$$

$$M = \left[\begin{array}{cccc} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{array} \right] \tag{8.9}$$

$$\begin{aligned} p(x) = & 3x^6 + 14x^5y + 590x^4y^2 \\ & + 19x^3y^3 - 12x^2y^4 - 12xy^5 \\ & + 2y^6 - a^3b^3 \end{aligned}$$

$$ralign = lalign \tag{8.10}$$

$$ralign12345 = lalign \tag{8.11}$$

$$ralign = 12345lalign \tag{8.12}$$

$x = y$	$w = z$	$a = b + c$
$2x = -y$	$3w = \frac{1}{2}z$	$a = b$
$-4 + 5x = 2 + y$	$w + 2 = -1 + w$	$ab = cb$

m	Modulo	$m > 0$
a	Multiplier	$0 < a < m$
c	Constant	$0 \leq c < m$
x_0	Initial Value	$0 \leq x_0 < m$

m	Modulo	$m > 0$
a	Multiplier	$0 < a < m$
c	Constant	$0 \leq c < m$
x_0	Initial Value	$0 \leq x_0 < m$

$eqn1 = fdsjalk$ (8.13)

$eqn2 = fdsa * 243$ (8.14)

eqn

$= eqn \quad (8.15)$

8.2 Functions

<code>\sin</code>	sin	<code>\cos</code>	cos	<code>\tan</code>	tan
<code>\arcsin</code>	arcsin	<code>\arccos</code>	arccos	<code>\arctan</code>	arctan
<code>\sec</code>	cos	<code>\csc</code>	csc	<code>\cot</code>	cot
<code>\sinh</code>	sinh	<code>\cosh</code>	cosh	<code>\tanh</code>	tanh
<code>\ln</code>	ln	<code>\log</code>	log	<code>\exp</code>	exp

Chapter 9

Citations, References, and Cross-References

This section will be showing off some of the different ways to include “citations” and “cross-references” within your document. Note that **cross-references** in L^AT_EX utilize `\ref` as a command, while one might think that this is short for reference this is not the case citation/references utilize the `\cite` command.

9.1 Cross-References

In L^AT_EX, references will “reference” a `\label{Reference:Label}` command. This section has the following command to define the Chapter:

```
\chapter{Citations/References, and Cross-References}\label{ch:citref}
```

By using `\ref{ch:citref}`, this allows you to insert a cross-reference that look like this 9. Now this by itself is not the most useful, to make it a bit better we should keep track of what we are cross-referencing, in this case a **Chapter**, and add this label in front of the cross-reference (`Chapter~\ref{sec:citref}`) and this will display like this: Chapter 9.

Note: *To ensure the reference is not split we need add a non-breaking space ($\sim$) to prevent $\LaTeX$ from accidentally adding a line-break between the label we added and the actual cross-reference.*

While using the `\ref` command, you might ask “*Why does $\LaTeX$ not just know what it is that I am referencing and insert that automatically in front of the reference?*” The answer is to provide more flexibility to the user. However, that being said, individuals have created a number of packages that work to enhance the workflow of adding these cross-references. Some of these are provided by the `hyperref` and `cleveref` packages. To include these packages add the following lines to the **bottom** of your preamble (order matters, `cleveref` needs to be after `hyperref`; and `hyperref` should be one of the last packages loaded):

```
\usepackage{hyperref}
\usepackage[nameinlink]{cleveref}
```

With these packages loaded we can now use the commands listed in [Table 9.1](#).¹

¹Note that because the floats are added where they are in the text this causes them to insert large amounts of white space because it only fits on the following page.

Table 9.1: Built-in, hyperref, and cleveref commands and outputs

Command	Output
built-in	
<code>\ref{}</code>	9.1
<code>\pageref{}</code>	66
hyperref	
<code>\autoref{}</code>	Table 9.1
cleveref	
<code>\cref{}</code>	Table 9.1
<code>\Cref{}</code>	Table 9.1
<code>\cref*{}</code>	Table 9.1
<code>\Cref*{}</code>	Table 9.1
<code>\cpageref{}</code>	Page 66
<code>\Cpageref{}</code>	Page 66
<code>\namecref{}</code>	Table
<code>\nameCref{}</code>	Table

Further, the `cleveref` also includes features that allows for the auto sorting and combining of references:

```
\Cref{fig:doubleImage,fig:singleImage,fig:tripleImage1,fig:quadImage}
```

Noting that there are **NO** spaces between the labels; this will produce: `????` and **Figs. A.2** and **A.4**. Allowing one to quickly and efficiently keep references up-to-date and consistent in their style. More examples of the use of the `cleveref` cross-referencing is found through the rest of this Chapter.

9.2 Citations/References

Citations are a lot easier than dealing with the cross-referencing. There are no additional packages required for citations, the built-in ones are feature-rich enough.

Single citations can be included with the `\cite{citationKey}` command, the one at the end of this sentence is created with the `\cite{TEST}` command [0]. Multiple citations can be included in a single cite command by adding commas in between the citation keys. The citation at the end of this sentence shows how to create more than one citation and how they are grouped together, it is created with the `\cite{testone,cite2,cite3,cite4,cite5}` command [0]. Finally this sentence shows how a gap in the citations is handled, this is created with the `\cite{testone,cite2,cite3,cite5}` command [0].

Finally, another useful citation command is the `\fullcite{citekey}` command that will print out the full citation. The following is produced using the `\fullcite{Aldrich2017}`:

D. R. Aldrich *et al.*, “Osm-classic: An optical imaging technique for accurately determining strain,” *SoftwareX*, vol. 6, pp. 225–230, 2017. DOI: [10.1016/J.SOFTX.2017.08.007](https://doi.org/10.1016/J.SOFTX.2017.08.007)

References

- [0] D. R. Aldrich, C. Ayranci, and D. S. Nobes, “Osm-classic: An optical imaging technique for accurately determining strain,” *SoftwareX*, vol. 6, pp. 225–230, 2017. DOI: [10.1016/J.SOFTX.2017.08.007](https://doi.org/10.1016/J.SOFTX.2017.08.007).
- [0] A. Thor, “Cited article,” *Journal Name*, 2020.
- [0] A. one, “Article 1,” *Journal 1*, 1989.
- [0] A. two, “Article 2,” *Journal 2*, 1900.
- [0] A. three, “Article 3,” *Journal 3*, 1990.
- [0] A. four, “Article 4,” *Journal 4*, 1990.

[0] A. five, “Article five,” *Journal five*, 1990.

Part III

Technical Writing and Reference Management

Chapter 10

TeXstudio: A Comprehensive IDE for $\text{\LaTeX}$

10.1 Introduction

While $\text{\LaTeX}$ documents can be written in any plain text editor, using a dedicated Integrated Development Environment (IDE) like TeXstudio significantly lowers the learning curve. TeXstudio is designed specifically for creating $\text{\LaTeX}$ documents, offering features like real-time syntax checking, integrated PDF viewing, and automated command completion. This chapter will provide an overview of TeXstudio's primary features and guide you through setting it up to work seamlessly with your thesis template.

10.2 Key Features of TeXstudio

TeXstudio stands out because it brings together all the tools needed for document preparation into a single interface, reducing the need to interact directly with the command line.

10.2.1 Interactive Syntax Checking

One of the most useful features for students is the “live” syntax checker. TeXstudio underlines errors in red and warnings in yellow as you type. This allows you to catch a missing bracket or a misspelled command immediately, rather than waiting until you try to compile the entire document.

10.2.2 Integrated PDF Viewer

TeXstudio includes a built-in PDF viewer that supports “SyncTeX.” This allows for two-way communication between your code and your output: if you right-click a sentence in the PDF, you can jump directly to the corresponding line in the L^AT_EX code, and vice versa.

10.2.3 Advanced Completion

When you begin typing a command (*e.g.*, `\cite{}`), TeXstudio provides a dropdown list of available citation keys from your bibliography. It also automatically suggests the required arguments for complex environments like tables and figures, which is a massive time-saver for a thesis.

10.2.4 Structure View

The side panel provides a hierarchical view of your document structure (Parts, Chapters, Sections). For a large thesis, this serves as an interactive Table of Contents, allowing you to jump between different chapters of your work with a single click.

10.3 Getting Started with TeXstudio

Setting up TeXstudio correctly is essential for ensuring that your thesis compiles without errors, especially when using the uAlberta template.

10.3.1 Installation

The software is available for Windows, macOS, and Linux. Visit the official website: <https://www.texstudio.org/> and download the installer for your operating system.

Note: *Ensure you have already installed a L^AT_EX distribution like MikTeX or TeXLive before installing TeXstudio, as the editor requires a compiler to function.*

10.3.2 The Main Interface

Upon launching TeXstudio, you will see the primary workspace as shown in [Figure 10.1](#).

10.3.3 Configuration for Your Thesis

To ensure the uAlberta template compiles correctly (especially if you are using specialized fonts or BibL^AT_EX), you should verify your build settings:

1. Go to ‘Options’ → ‘Configure TeXstudio’.
2. Under the ‘Build’ tab, ensure the ‘Default Compiler’ is set to **Pd^fLaTeX**.
3. Ensure the ‘Default Bibliography Tool’ is set to **Biber**.

10.4 Efficient Writing Workflows

Beyond just typing text, TeXstudio has several tools that make the technical side of thesis writing easier.

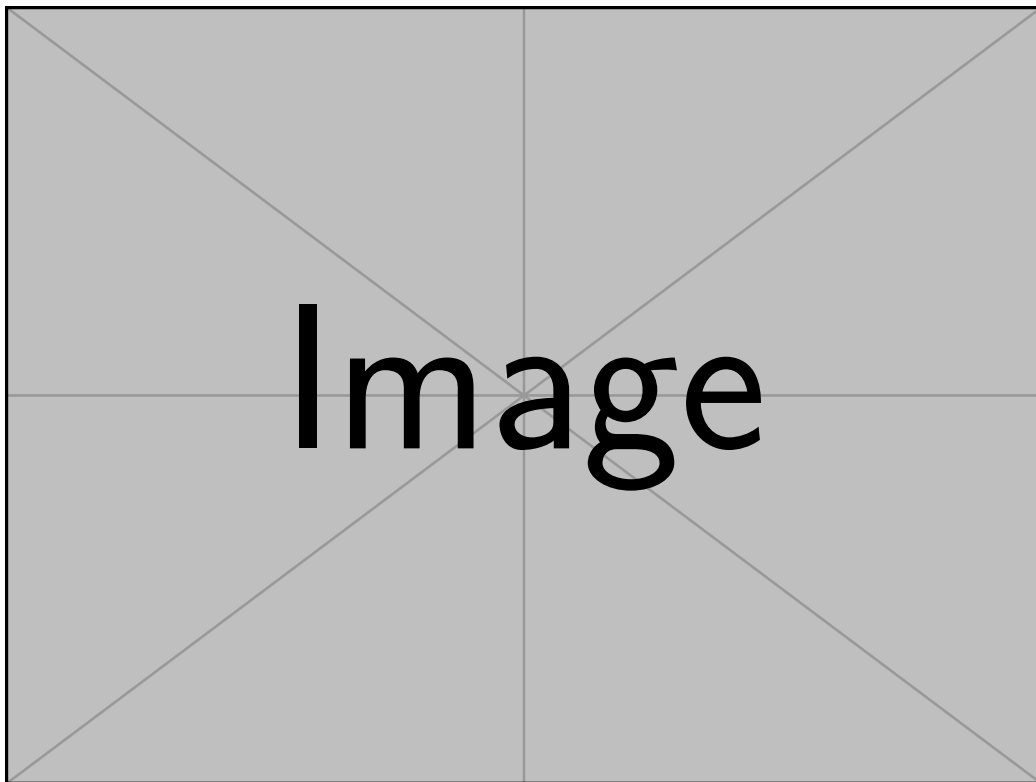


Figure 10.1: TeXstudio Main Interface showing editor, structure, and PDF viewer.

10.4.1 Using the Bibliography with JabRef

If you have your JabRef library open and saved as a `.bib` file in your project folder, TeXstudio will automatically “read” it. When you type `\cite{`, a list of your references will appear. You can search this list by author or title directly within the editor.

10.4.2 Managing Large Documents

Since a thesis is usually split into multiple files (one for each chapter), you must tell TeXstudio which file is the “Root” file. This can be done in one of two ways.

The first is to use “Magic Comments” at the top of each separate file. The format of this comment (which should be the first line of the file) should be: `% !TeX root = ../ualberta.tex`.

The second way is to set the root in TeXstudio.

1. Open your main thesis file (*e.g.*, `ualberta.tex`).
2. Go to ‘Options’ → ‘Root Document’ → ‘Set Current Document as Explicit Root’.

This ensures that whenever you press “Compile” while working on a sub-chapter file, TeXstudio knows to compile the entire thesis starting from the main file.

10.4.3 Mathematical Assistance

For the complex equations required in engineering and technical theses, the ‘Mathematical Symbols’ panel on the left provides a clickable library of Greek letters, operators, and relations. Hovering over a symbol will even show you the required L^AT_EX command.

Chapter 11

JabRef: Managing Bibliographies Efficiently

11.1 Introduction

JabRef stands as a powerful tool for researchers and academics engaged in scholarly writing. JabRef offers a robust solution for bibliography management, including a number of features to ensure that you are not only able to organize your references but keep track of progress, and notes on each reference. This chapter aims to provide an review of JabRef, including its myriad features that I find particularly useful, and to guide you through its implementation in writing your thesis.

11.2 Key Features of JabRef

JabRef, with its versatile features, emerges as an indispensable tool for bibliography/reference management. Delving deeper into its functionalities reveals a wealth of tools designed to streamline the often cumbersome process of handling references.

11.2.1 BibTeX Compatibility

JabRef's commitment to the BibTeX format is a testament to its roots in the L^AT_EX ecosystem. This compatibility ensures a seamless integration between the reference management process and the L^AT_EX document preparation workflow. Users can easily export and import BibTeX files, facilitating collaboration and compatibility across various platforms.

11.2.2 Reference Import

The capability to import references directly from online databases and journal websites significantly accelerates the reference collection process. JabRef supports various import formats, allowing users to effortlessly populate their databases with accurate and structured reference information. This feature is particularly valuable for researchers dealing with large/extensive bibliographies.

11.2.3 Customizable Entry Types

The flexibility offered by customizable entry types allows users to categorize references based on the nature of the source. Whether it's a book, article, conference proceeding, or any other reference type, JabRef accommodates diverse sources, ensuring a well-organized and easily navigable bibliography.

11.2.4 Search and Filter

The ability to efficiently search and filter references is a hallmark of JabRef's usability. Researchers dealing with extensive databases will appreciate the quick and precise retrieval of references based on author names, titles, keywords, or any other criteria. This feature is crucial for maintaining order in a rapidly growing bibliography.

11.2.5 Grouping

JabRef’s grouping functionality provides a systematic approach to organizing references. Users can create custom groups to categorize references based on themes, projects, or any other criteria. This feature is especially useful for large research projects where a systematic organization of references is essential for maintaining clarity and coherence.

11.2.6 Integration with L^AT_EX

The seamless integration of JabRef with L^AT_EX editors fortifies the synergy between bibliography management and document preparation. This integration minimizes the manual effort required for citation insertion and ensures consistency between the bibliography and the in-text citations. Users can easily copy citation keys from JabRef and paste them directly into their L^AT_EX documents.

11.3 Getting Started with JabRef

Now that we’ve outlined the key features of JabRef, let’s embark on a comprehensive guide on how to get started with JabRef. This step-by-step walkthrough will cover everything from installation to creating a new bibliography and populating it with references.

11.3.1 Installation

The initial step in utilizing JabRef is to install the software on your system. For all users, regardless of OS, the easiest way to download JabRef is to visit their website: <https://www.jabref.org/>. Once there select ‘Download’ from the navigation bar, and press the “Download JabRef” button. This will take you to the FossHub page where you can select the appropriate version for your OS and download and install it.

11.3.2 Creating a New Bibliography

Once JabRef is successfully installed, launch the application. When the program loads you will be faced with a window that looks like the one shown in [Figure 11.1](#). Now that the program is

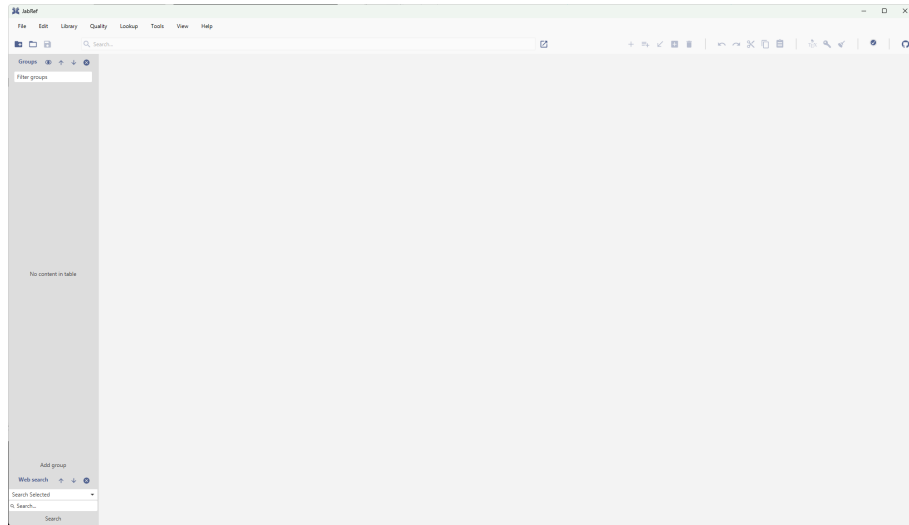


Figure 11.1: JabRef Main Window.

open, to create a new bibliography:

1. Click on ‘File’ → ‘New Library’.

To save the database:

1. Click on ‘File’ → ‘Save Library’.
2. Choose an appropriate and location.
3. Click ‘Save’.

Congratulations! You’ve initiated your bibliography using JabRef. Now that we have this created, the next step is to add references to the database.

11.3.3 Adding References

JabRef offers multiple avenues for adding references to your database. Some of the methods are generally more useful than others but we will go over a few that you are likely to use:

11.3.3.1 Web Search

JabRef's integrated web search (see [Figure 11.2](#)) feature simplifies the process of importing references from online sources. This is by-far the easiest way to enter a reference.

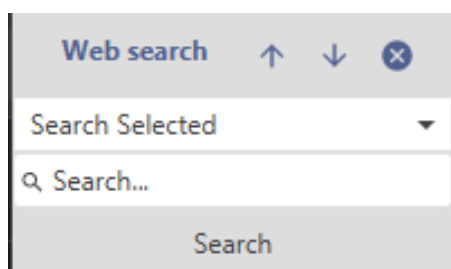


Figure 11.2: JabRef Web Search Tool.

1. Click on 'Web Search'.
2. Search for the desired reference using the integrated search feature.
3. Select the reference all the references you wish to import, as shown in [Figure 11.3](#).
4. Click 'Import entries' to import the selected entries.
5. The references are added to your library.

Note: The Web Search tool by default uses a general search, however, a specific database can be chosen as well using the drop down arrow next to "Search Selected".

11.3.3.2 Manual Entry

Manually adding a reference can be done in a 'manual' and 'automatic' way. When adding a new entry you will be faced with the option to select an entry type or to enter an ID (DOI, ArXiv,

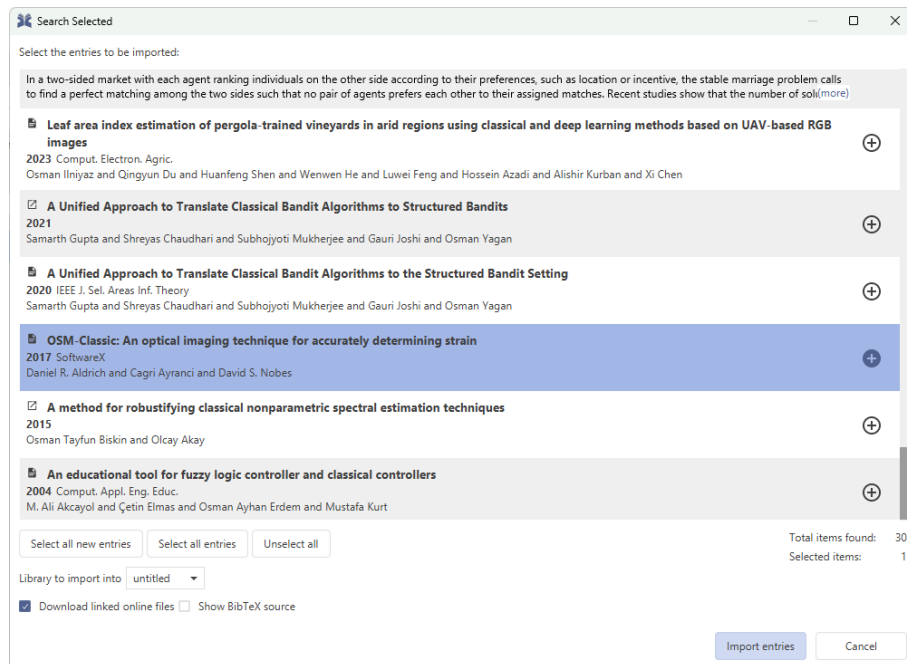
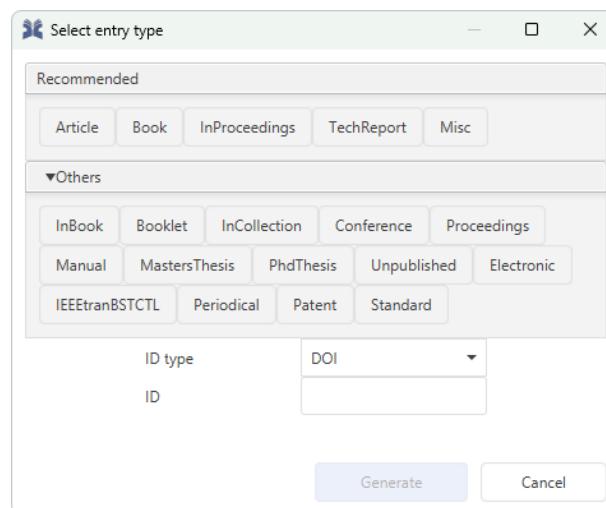


Figure 11.3: Example Web Search Results for “OSM-Classic”.

ISBN, *etc.*). If you enter an ID, the information for the reference will be automatically pulled from the internet. Otherwise to manually enter all the information for a reference:

1. Click ‘Library’ → ‘New entry’ or use the shortcut ‘Ctrl + N’ and this will show the following window.



2. Choose the entry type (*e.g.*, article, book, inproceedings).

3. Fill in the required fields like author, title, journal, *etc.*

By following these steps, you can efficiently populate your JabRef database with the necessary references.

11.3.4 Organizing References

Effectively organizing references is essential for a streamlined bibliography. JabRef's grouping feature allows you to categorize references based on your preferences:

1. On the left panel, select 'Add Group'.
2. Give the group an appropriate name.
3. Optionally you can add a Description, Icon, Colour, *etc.*

To assign a reference to a group:

1. Select the Reference(s) from the centre list.
2. Drag them to the group on the left of the screen.

Note: *Groups can even be nested into groups to provide more levels of organization.*

Organizing references into groups enhances accessibility and facilitates a more systematic approach to bibliography management.

11.4 Exploring Advanced Features of JabRef

JabRef's capabilities extend beyond the basics covered in the previous sections. In this section, we will explore some of the advanced features that enhance the efficiency and effectiveness of JabRef as a reference manager.

11.4.1 Quality Assurance: Checking and Correcting Entries

Ensuring the accuracy and completeness of references is crucial. JabRef provides tools for quality assurance, allowing users to check and correct entries.

To check for duplicate entries:

1. Click on ‘Quality’ → ‘Find duplicates’.
2. JabRef will identify and display duplicate entries.

To correct entries:

1. Click on ‘Quality’ → ‘Cleanup entries’.
2. JabRef will provide some useful option to ensure conformity within the different references.

This includes renaming Linked PDF’s to match the standard of “`CitationKey - Title`”.

These quality assurance features contribute to maintaining a clean and error-free bibliography.

11.4.2 Managing PDFs and File Links

JabRef facilitates the management of associated PDF’s and file links, offering a consolidated approach to reference and document management.

To link a PDF or file:

1. Open the entry editor for a reference.
2. Click on ‘General’ and use the ‘PDF’ or ‘File’ field to link the document.

This integration helps to streamline the retrieval of PDF’s or other associated documents directly from JabRef. Further, this allows JabRef to keep track of the comments and highlights in a single place (see [Figure 11.4](#)). These annotations can be found by selecting the entry, and selecting the ‘File annotations’ tab.

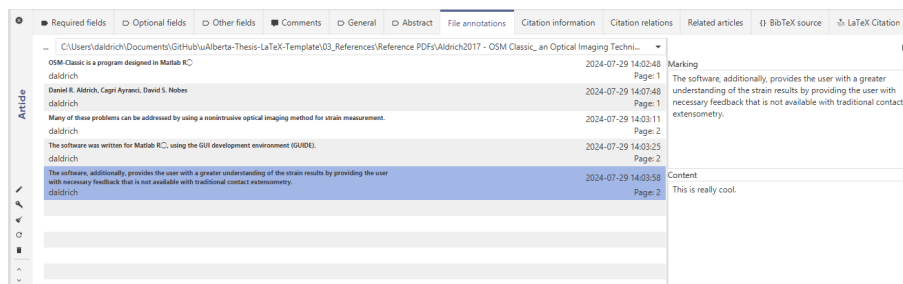


Figure 11.4: Showcase of the file annotations in JabRef.

11.4.3 Additional Information

JabRef keeps track of a lot of information and can even help you with your research. Some additional information JabRef provides includes:

- Citation information
- Citation relationships (what the reference cites and who has cited the reference). This further lets you open the links to the reference's source, or even add these references directly to your library.
- If one right clicks an entry you are provided the following options:
 - Rank - rank the reference with one to five stars.
 - Toggle Relevance - add a marker to show this is a relevant source.
 - Priority - rank items as low, medium, or high priority.
 - Read Status - set the status to read or skimmed.

Chapter 12

Notepad++: A Lightweight Text Editor for Advanced Workflows

12.1 Introduction

While dedicated IDEs like TeXstudio are highly recommended for writing the bulk of your thesis, there is undeniable value in having a fast, lightweight text editor in your toolkit. Notepad++ is a free, open-source source code editor that supports several languages, including L^AT_EX.

Unlike TeXstudio, Notepad++ will not hold your hand with integrated PDF viewers or built-in mathematical symbol libraries. However, its speed, robust search capabilities, and flexibility make it an excellent secondary tool for managing raw code, editing configuration files, or cleaning up messy bibliography data.

12.2 Key Features of Notepad++

Notepad++ strips away the heavy graphical interfaces of full IDEs, focusing entirely on efficient text manipulation.

12.2.1 Speed and Low Resource Usage

Because it is a plain text editor at its core, Notepad++ opens almost instantly, even when loading massive text files or raw data sets. This makes it ideal for quick edits when you do not want to wait for a full IDE to initialize.

12.2.2 Advanced Search and Replace

Perhaps the most powerful feature of Notepad++ is its “Find in Files” capability. If you realize you have used the wrong terminology or need to change a variable name across all twenty chapters of your thesis, Notepad++ can search your entire project folder and replace the text in one click. It also fully supports Regular Expressions (Regex) for highly complex search patterns.

12.2.3 Syntax Highlighting

Out of the box, Notepad++ recognizes `.tex` and `.bib` files and will apply basic syntax highlighting to your $\text{\LaTeX}$ commands, making the raw code much easier to read than in standard Windows Notepad.

12.2.4 Plugin Ecosystem

Through the built-in Plugin Admin, you can extend Notepad++’s functionality. Plugins like *NppExec* allow you to send commands directly to your computer’s command line, meaning you can set up a custom shortcut to compile your $\text{\LaTeX}$ document directly from the editor.

12.3 Getting Started with Notepad++

Setting up Notepad++ is quick and straightforward.

12.3.1 Installation

Notepad++ is available exclusively for Windows. To install the software, visit the official website: <https://notepad-plus-plus.org/>. Navigate to the ‘Download’ section, select the latest release, and download the installer. Follow the standard Windows installation prompts.

12.3.2 The Main Interface

Upon launching, you will see a clean, tabbed interface as shown in **Figure 12.1**.

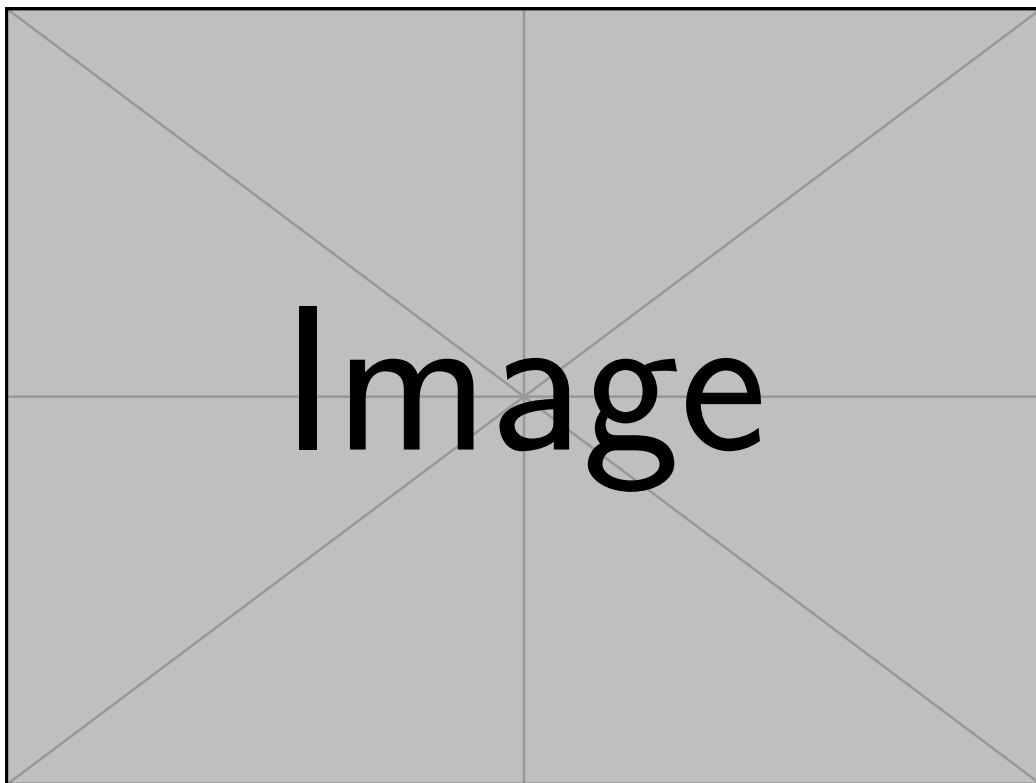


Figure 12.1: Notepad++ Main Interface showing a $\text{\LaTeX}$ document with syntax highlighting.

12.3.3 Compiling $\text{\LaTeX}$ from Notepad++ (Advanced)

If you wish to use Notepad++ as your primary editor, you will need a way to compile your document into a PDF. This requires the *NppExec* plugin:

1. Go to ‘Plugins’ → ‘Plugins Admin’.
2. Search for ‘NppExec’, check the box, and click ‘Install’. Notepad++ will restart.
3. Press ‘F6’ to open the Execute dialog.
4. Enter the following commands to compile your document:

```
cd "$(CURRENT_DIRECTORY)"
```

```
latexmk -pdf -silent -synctex=1 "$(FILE_NAME)"
```

5. Save this script for future use.

Note: *For most thesis writers, it is far easier to use TeXstudio for compiling and reserve Notepad++ for quick text edits.*

12.4 Notepad++ in a Thesis Workflow

You might wonder why you need Notepad++ if you already have TeXstudio. Here are a few scenarios where a lightweight text editor shines:

12.4.1 Editing Style (.sty) and Class (.cls) Files

If you need to make a minor tweak to the uAlberta thesis template’s underlying code (such as adjusting a margin or fixing a title page layout), opening the .cls file in Notepad++ is often cleaner and safer than loading it into your main IDE.

12.4.2 Cleaning up BibTeX Files

Sometimes, exporting references from online databases results in messy .bib files with broken characters or formatting errors. Opening your bibliography database in Notepad++ allows you to view the raw text without any software trying to “interpret” it, making it easier to spot and delete corrupted lines.

12.4.3 Managing External Scripts

If your research involves writing Python or other programming scripts for data analysis or generating complex TikZ plots, Notepad++ allows you to edit those external scripts side-by-side with your L^AT_EX code in a lightweight environment.

Part IV

Data Analysis and Programming

Chapter 13

GNU Octave: Open-Source Numerical Computation and Plotting

13.1 Introduction

While L^AT_EX is an unparalleled tool for typesetting your thesis, it is not designed to crunch numbers or analyze raw data. For that, you need a numerical computation environment. GNU Octave is a high-level programming language primarily intended for numerical computations, and it is largely compatible with MATLAB. This chapter will introduce you to GNU Octave and explain how to use it to process your research data and generate publication-quality figures that integrate perfectly into your L^AT_EX document.

13.2 Key Features of GNU Octave

GNU Octave provides a powerful suite of tools for researchers, particularly those in engineering and the hard sciences, without the steep licensing fees associated with commercial software.

13.2.1 MATLAB Compatibility

The defining feature of Octave is its syntax compatibility with MATLAB. If you have written scripts or functions for MATLAB during your coursework or laboratory research, they will almost always run in Octave without modification. This makes it an ideal tool for working on your thesis from home or on a personal machine.

13.2.2 Advanced Data Visualization

A thesis requires clear, precise, and professional plots. Octave excels at 2D and 3D plotting. More importantly for our purposes, it allows you to export these plots as scalable vector graphics (such as PDF or EPS files), ensuring that your graphs never become pixelated or blurry when compiled into your $\text{\LaTeX}$ document.

13.2.3 Extensive Package Ecosystem

Similar to MATLAB toolboxes, Octave has a repository of community-contributed packages (Octave Forge). Whether you need to analyze hydraulic system responses, process control signals, or perform complex statistical modelling, there is likely a package available to simplify the task.

13.3 Getting Started with GNU Octave

Getting Octave running on your machine is a straightforward process.

13.3.1 Installation

GNU Octave is free and available for Windows, macOS, and Linux. To download the software, visit the official website: <https://octave.org/>. Navigate to the ‘Download’ section, select the installer for your operating system, and follow the standard installation prompts.

Note: *During installation on Windows, ensure you select the GUI (Graphical User Interface) version, rather than just the command-line interface, as it is much easier for managing scripts and plots.*

13.3.2 The Main Interface

When you launch the Octave GUI, you will be presented with a workspace that integrates several important panels, as shown in [Figure 13.1](#).

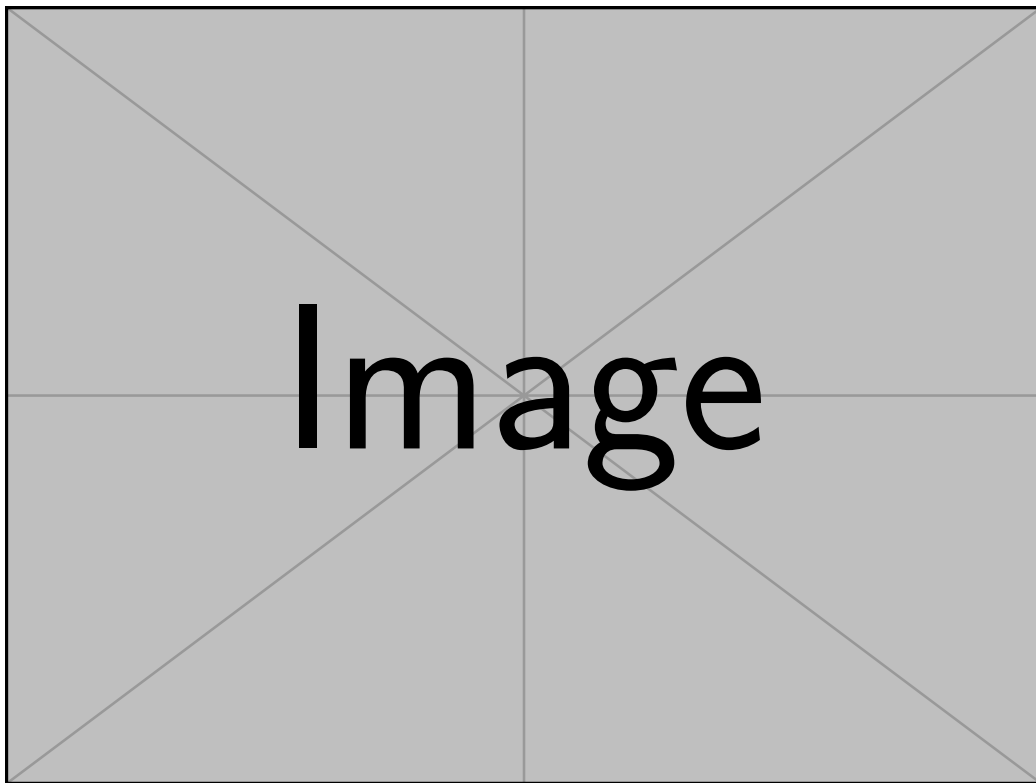


Figure 13.1: GNU Octave GUI showing the Command Window, Workspace, and Editor.

The main areas include:

- **Command Window:** Where you can type commands for immediate execution.
- **Workspace:** Displays all currently active variables and their values.
- **Editor:** A built-in text editor for writing and saving longer scripts (`.m` files).

13.4 Octave in a L^AT_EX Thesis Workflow

The most critical interaction between GNU Octave and L^AT_EX is the generation of figures. Copying and pasting screenshots of your plots is highly discouraged in a thesis. Instead, you should script your plots to export automatically.

13.4.1 Exporting Vector Graphics

When you generate a plot in Octave, you want to save it in a format that L^AT_EX handles well natively, such as PDF. This ensures the text and lines in your graph scale perfectly.

In your Octave script, after you have plotted your data and added your labels, you can use the `print` command to save the figure:

```
print -dpdf my_thesis_plot.pdf
```

This file can then be directly included in your thesis using the standard `\includegraphics` command or the custom command `\insertimage`¹ within a `figure` environment.

13.4.2 Consistent Formatting

To make your thesis look cohesive, the font sizes and styles in your Octave plots should roughly match the text in your L^AT_EX document. You can control this directly in your Octave scripts by specifying font properties before printing:

```
set(gca, 'fontsize', 12, 'fontname', 'Times')
```

¹The `\insertimage` just wraps the `\includegraphics` command by automatically adding the image directory to the filename.

i.e., `\insertimage{filename}` is equivalent to `\includegraphics{./99_Inclusions/Images/filename}`.

13.4.3 Advanced: TikZ Export

For the highest possible quality, advanced L^AT_EX users sometimes use external scripts (like `matlab2tikz`) within Octave to export their plots directly into L^AT_EX code (using the `PGFPlots` or `TikZ` packages). While this creates perfectly matched fonts and styles, it significantly increases compile time and is generally only recommended for simple 2D plots.

13.4.4 Advanced: TikZ/PGFPlots Export

For the highest possible quality and perfect font matching with your thesis template, you can export your Octave plots directly into L^AT_EX code rather than saving them as PDFs.

GNU Octave has native support for this via the `print` command using the `-dtikz` device:

```
print -dtikz my_plot.tex
```

This will generate a `.tex` file containing raw PGF drawing commands, which you can then include in your document using `\input{./99_Inclusions/plots/my_plot.tex}` or using the custom command `\insertplot{my\_plot.tex}`.

Note: While the native export is convenient, many advanced users prefer to use an open-source script called `matlab2tikz`. Rather than generating low-level drawing commands, `matlab2tikz` extracts your data and generates a clean `pgfplots` environment. This makes the resulting code much easier to read and allows you to adjust the plot's scale, line weights, and axis labels directly within TeXstudio without having to regenerate the figure in Octave.

Chapter 14

Git and GitHub: Robust Version Control for Your Thesis

14.1 Introduction

Writing a thesis is a marathon, and over the course of months or years, your document will undergo hundreds of revisions. Relying on the traditional method of saving files as `thesis_v1`, `thesis_v2`, and `thesis_final` quickly leads to a chaotic and unmanageable project folder.

To solve this, we use **Git**, an industry-standard version control system, paired with **GitHub**, a cloud-based hosting service. Just as physical equipment designs rely on rigorous revision control to track changes and prevent manufacturing errors, a complex $\text{\LaTeX}$ document requires a systematic approach to track document history, allow for safe experimentation, and prevent catastrophic data loss.

14.2 Understanding Git vs. GitHub

It is important to distinguish between these two interconnected tools:

- **Git:** The underlying software that runs locally on your computer. It acts as a tracking system, recording exactly what was changed, when, and by whom, line-by-line.
- **GitHub:** The cloud-based platform where you upload (or “push”) your Git repository. It serves as a secure off-site backup and a hub for collaboration if your supervisor wishes to review your source code.

14.3 Key Features of Version Control

Implementing version control might seem like an extra step, but its benefits for a large document are invaluable.

14.3.1 Complete Document History and Rollback

Every time you save a significant chunk of work, you create a “commit” in Git. If you delete a section of your literature review and realize three weeks later that you actually need it, you can easily browse your Git history and restore that specific text without affecting the rest of your current document.

14.3.2 Secure Off-site Backup

Hard drives fail, and laptops get stolen. By regularly pushing your commits to a private GitHub repository, you ensure that an up-to-date, trackable copy of your thesis is always safe in the cloud.

14.3.3 Safe Experimentation (Branching)

Suppose you want to completely restructure your methodology chapter, but you are worried about breaking the uAlberta template. Git allows you to create a “branch”—an isolated copy of

your thesis. You can make all the changes you want on this branch. If it works, you merge it back into your main document. If it breaks, you simply delete the branch and return to your stable main version.

14.4 Getting Started with GitHub Desktop

While Git can be run entirely from the command line, the easiest way to manage your thesis is through a graphical interface like GitHub Desktop.

14.4.1 Installation and Setup

1. Create a free account on <https://github.com/>.
2. Download and install **GitHub Desktop** from <https://desktop.github.com/>.
3. Open the application and sign in with your GitHub credentials.

14.4.2 Creating Your Thesis Repository

A “repository” (or repo) is simply the main folder containing all your thesis files.

1. In GitHub Desktop, go to ‘File’ → ‘New repository’.
2. Give it a name (*e.g.*, `Main_Thesis_Smith`).
3. Choose the local path on your computer where your L^AT_EX files are currently stored.
4. Check the box to keep the repository **Private** (you do not want your unpublished research available to the public).
5. Click ‘Create repository’ and then click ‘Publish repository’ to send the initial backup to the cloud.

14.5 The L^AT_EX Workflow with Git

Using Git with L^AT_EX requires one specific, critical configuration: the `.gitignore` file.

14.5.1 Ignoring Auxiliary Files

When you compile a document in TeXstudio or Notepad++, the compiler generates a multitude of temporary files (`.aux`, `.log`, `.out`, `.toc`, `.synctex.gz`, etc.). **You should never track these files in Git.** Tracking them creates massive amounts of useless history and constant “fake” file changes.

Git handles this using a file called `.gitignore`.

1. In the root folder of your thesis, create a plain text file named exactly `.gitignore` (note the leading period).
2. Open it and add the file extensions you wish to ignore. For a standard L^AT_EX setup, you can copy a pre-made template from [GitHub’s official TeX gitignore list](#).

Once saved, GitHub Desktop will only track your core files (`.tex`, `.bib`, and your image/plot files).

14.5.2 The Daily Routine: Commit and Push

Your workflow when writing should look like this:

- **Write:** Work on a section of your thesis in TeXstudio.
- **Review:** Open GitHub Desktop. It will highlight the exact lines of text you added or modified.
- **Commit:** In the bottom left corner, write a short, descriptive summary (*e.g.*, “*Drafted subsection on thesis topic A*”) and click ‘Commit to main’.
- **Push:** Click ‘Push origin’ at the top of the screen to send your new commit to the cloud.

Note: *Get into the habit of committing your work logically. Do not wait until the end of the week to make one massive commit called “Updates.” Commit every time you finish a section, fix a complex table, or update a series of figures.*

Chapter 15

Calcpad: Transparent Engineering Calculations

15.1 Introduction

In many technical and engineering theses, simply stating a final numerical result is insufficient; your reviewers and defence committee will want to see the underlying design calculations. While Microsoft Excel is ubiquitous in engineering, it is notoriously poor for academic documentation because formulas are hidden within cells, making peer review incredibly tedious.

Calcpad is a free, programmable software designed specifically for mathematical and engineering calculations. It takes your raw formulas, variables, and units, and automatically generates beautifully formatted, step-by-step calculation reports. This chapter will explore how Calcpad can replace cumbersome spreadsheets and provide transparent, verifiable math for your thesis appendices.

15.2 Key Features of Calcpad

Calcpad operates similarly to a programming script, but its output is purely focused on readable mathematics.

15.2.1 Transparent Variable Substitution

The most powerful feature of Calcpad is how it displays results. When you define an equation, Calcpad does not just output the final number. It outputs the algebraic formula, the formula with the substituted numerical values, and the final result (*e.g.*, $A = b \cdot h = 2 \text{ m} \cdot 3 \text{ m} = 6 \text{ m}^2$). This level of transparency is exactly what a thesis committee expects to see in a sample calculation appendix.

15.2.2 Intelligent Unit Management

Engineering calculations are prone to unit conversion errors. Calcpad has a built-in, comprehensive library of units (SI, Imperial, and US Customary). If you attempt to add a force in Newtons to a pressure in Pascals, Calcpad will flag the error. Furthermore, it automatically handles complex conversions behind the scenes, ensuring your final results are dimensionally accurate.

15.2.3 HTML and PDF Output

Calcpad parses your scripts and renders the output as a clean HTML document using MathJax (the same underlying rendering technology used by many $\text{\LaTeX}$ web viewers). These reports can be directly printed to high-quality PDF files, making them incredibly easy to append to your main thesis document.

15.3 Getting Started with Calcpad

Setting up Calcpad is quick and requires minimal system resources.

15.3.1 Installation

Calcpad is a free, open-source application available for Windows. To install the software:

1. Visit the official website: <https://calcpad.eu/> or their GitHub repository.
2. Download the latest `.exe` installer for Windows (either the x64 or x86 version depending on your system).
3. Follow the standard installation prompts.

15.3.2 The Main Interface

Upon launching Calcpad, you will see a split-screen interface as shown in [Figure 15.1](#).

The interface is divided into two primary sections:

- **Code Editor (Left):** Where you type your variables, formulas, and comments using Calcpad's simple syntax.
- **Output Window (Right):** A real-time preview of your formatted calculations. Every time you press 'F5' (or click calculate), this window updates.

15.4 Integrating Calcpad into Your Thesis Workflow

While Calcpad formats its own math beautifully, it is a separate program from your L^AT_EX IDE.

Here is how best to use them together.

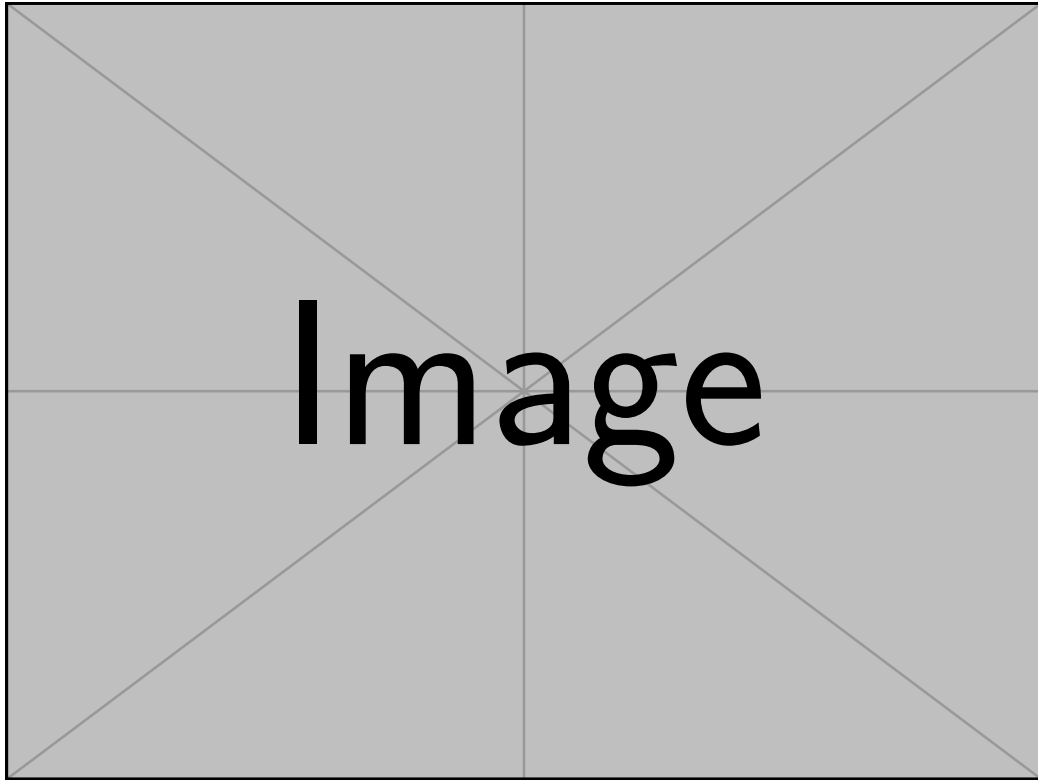


Figure 15.1: Calcpad Main Interface showing the code editor on the left and the rendered HTML output on the right.

15.4.1 Generating Calculation Appendices

Rather than re-typing pages of step-by-step math into L^AT_EX (which is highly prone to transcription errors), you can write your sample calculations in Calcpad and export them as a PDF.

Once you have saved your Calcpad PDF (*e.g.*, `Appendix_A_Sample_Calcs.pdf`), you can seamlessly merge it into your thesis using the `pdfpages` package in L^AT_EX:

```
\includepdf[pages=-, pagecommand=\  
thispagestyle{plain}]{Appendix_A_Hydraulic_Calcs.pdf}
```

This command inserts the external PDF directly into your thesis while maintaining your uAlberta template’s page numbering at the bottom.

15.4.2 A L^AT_EX Scratchpad

Writing complex, multi-line equations in L^AT_EX (using `align` or `split` environments) can be visually confusing in the raw code. Many engineers use Calcpad as a “scratchpad.” You can define your complex formula in Calcpad, ensure the math is physically correct and the units balance, and then carefully transcribe the final, verified algebraic equation into your main thesis chapters.

Note: *Calcpad supports adding plain text and HTML formatting alongside your equations. Use this feature to add descriptive headings and explanations between your formulas so that your appended PDF reads like a professional engineering report rather than just a list of numbers.*

Part V

Images and Technical Figures

Chapter 16

ImageJ (Fiji): Scientific Image Processing

16.1 Introduction

A technical thesis often requires the inclusion of experimental photographs, such as microscope micrographs, material cross-sections, or thermal imaging. While standard photo editors like Adobe Photoshop or Microsoft Paint can alter images, they are not designed for scientific rigor and can easily corrupt your data or degrade image quality.

ImageJ is a public domain, Java-based image processing program developed at the National Institutes of Health (NIH). It is the gold standard for scientific image analysis. This chapter will cover how to use ImageJ to analyze your experimental images, extract quantitative data, and properly format your pictures for high-quality printing in your $\text{L}^{\text{A}}\text{T}_{\text{E}}\text{X}$ thesis.

16.2 Key Features of ImageJ

ImageJ is built for scientists and engineers, meaning its tools prioritize measurable data over aesthetics.

16.2.1 Spatial Calibration and Measurement

Unlike a standard photo viewer that only understands “pixels,” ImageJ allows you to spatially calibrate your images. If you know that 500 pixels in your microscope image equates to 10 micrometers (μm), you can set this scale in ImageJ. Once calibrated, you can use the measurement tools to accurately calculate areas, perimeters, particle counts, and feature lengths in real-world units.

16.2.2 Scientific Scale Bars

A microscope or macro photograph without a scale bar is scientifically useless. Because ImageJ understands your spatial calibration, it can automatically generate and stamp highly accurate, customizable scale bars directly onto your images.

16.2.3 Non-Destructive Processing

When adjusting the brightness and contrast of a dark image so that it prints clearly in your thesis, ImageJ modifies the display lookup table (LUT) rather than permanently destroying the underlying pixel data, ensuring your raw data remains mathematically intact.

16.3 Getting Started with Fiji (ImageJ)

While you can download the base version of ImageJ, it is highly recommended to download **Fiji** (which stands for *Fiji Is Just ImageJ*). Fiji is a “batteries-included” distribution of ImageJ that comes pre-packaged with all the most useful scientific plugins and automatic updaters.

16.3.1 Installation

Fiji is completely free and available for Windows, macOS, and Linux.

1. Visit the official website: <https://imagej.net/software/fiji/>.
2. Download the appropriate package for your operating system.
3. Fiji does not require a traditional installer on Windows; simply extract the downloaded .zip file into a safe directory (like C:\Programs\Fiji) and run `fiji-windows-x64.exe`.

16.3.2 The Main Interface

When you launch Fiji, you will notice that it does not open a massive, screen-filling workspace. Instead, it opens a small, floating toolbar as shown in **Figure 16.1**.

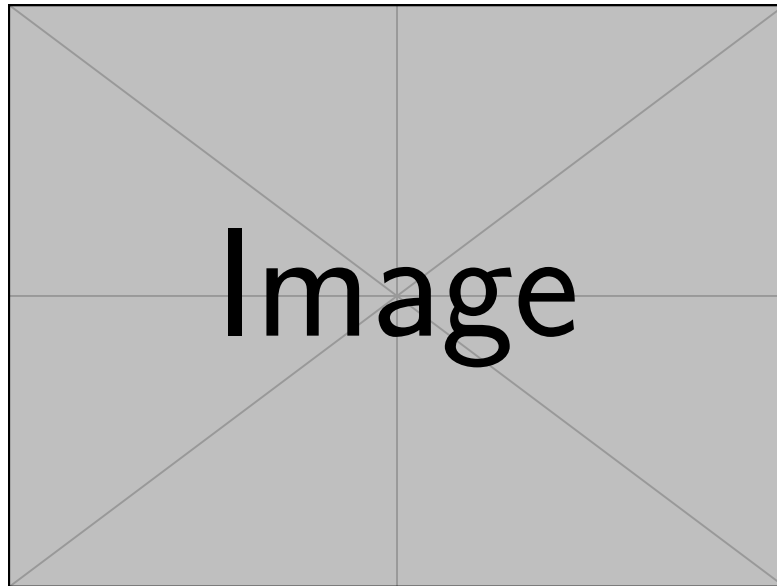


Figure 16.1: The floating Fiji (ImageJ) main toolbar.

When you drag and drop an image onto this toolbar, the image will open in its own separate window. All tools, filters, and measurement functions selected from the toolbar will apply to the currently active image window.

16.4 ImageJ in a L^AT_EX Workflow

The primary goal of using ImageJ in your thesis workflow is to prepare your experimental photographs so they compile cleanly and look professional in your final PDF.

16.4.1 Adding a Scale Bar

Before inserting an experimental image into your thesis, you must add a scale bar.

1. Open your image in Fiji.
2. Calibrate the image using ‘Analyze’ *rightarrow* ‘Set Scale’. Input the known distance and unit (*e.g.*, 10, μm).
3. Go to ‘Analyze’ *rightarrow* ‘Tools’ *rightarrow* ‘Scale Bar’.
4. Adjust the width, height, font size, and color (usually black or white depending on the background). Ensure the ‘Hide Text’ option is unchecked so the unit is visible.
5. Click ‘OK’ to stamp the scale bar onto the image.

16.4.2 Exporting the Final Image

L^AT_EX and the uAlberta template compile best with high-resolution, lossless image formats.

Never save your scientific images as .jpg files. The JPEG format uses “lossy” compression, which creates blurry artifacts around sharp edges like scale bars, text, and fine material features.

Instead, once your image is processed, use ‘File’ *rightarrow* ‘Save As’:

- **PNG (.png):** The best choice for L^AT_EX. It provides lossless compression, meaning your image retains 100% of its quality and the scale bar text remains perfectly crisp, while keeping the file size reasonable.
- **TIFF (.tif):** The archival standard for scientific images. TIFFs are completely uncompressed. You can use these in L^AT_EX, but be warned that if you have dozens of TIFFs, your

final thesis PDF file size may become massive.

Once exported as a PNG, you can comfortably insert the image into your document using the standard `\includegraphics` or the custom `\insertimage` command within a `figure` environment.

Chapter 17

GIMP: Preparing Illustrative and Equipment Photographs

17.1 Introduction

While ImageJ is the strict standard for analyzing scientific data and micrographs, a thesis also requires standard photography. You will often need to include pictures of your experimental setup, custom-built test rigs, or field equipment.

GIMP (GNU Image Manipulation Program) is a free, open-source raster graphics editor that serves as a highly capable alternative to Adobe Photoshop. This chapter will cover how to use GIMP to clean up distracting backgrounds, correct lighting, and ensure your illustrative photographs are formatted at the correct resolution for high-quality printing in your L^AT_EX document.

WARNING

You must never use GIMP’s advanced cloning, healing, or manipulation tools on scientific data (such as altering a material fracture surface or removing an “outlier” particle from a microscope image). GIMP should be strictly reserved for illustrative photographs where the physical appearance of the equipment, rather than extractable data, is the focus.

17.2 Key Features of GIMP

GIMP provides a comprehensive suite of tools for photo retouching and image composition.

17.2.1 Layers and Masks

Unlike basic photo viewers, GIMP utilizes a layer-based workflow. This allows you to stack edits on top of your original image without destroying it. More importantly, it allows you to use “Layer Masks” to hide distracting backgrounds. If you take a photo of a piece of equipment in a cluttered workshop, you can use masks to isolate the equipment and place it on a clean, professional white background for your thesis.

17.2.2 Perspective and Lens Correction

When taking photos of control panels or large equipment in tight laboratory spaces, the image is often skewed by lens distortion or captured at an angle. GIMP includes powerful perspective transformation tools that allow you to “square up” your photos, making them look as though they were taken perfectly straight-on.

17.2.3 Color and Lighting Adjustments

Lab environments often have terrible fluorescent lighting that washes out the contrast in your photos. GIMP’s “Levels” and “Curves” tools allow you to correct the white balance and pull detail out of dark shadows, ensuring your equipment is clearly visible when printed on paper.

17.3 Getting Started with GIMP

GIMP is a fully-featured program with a layout that will be familiar to anyone who has used legacy versions of Photoshop.

17.3.1 Installation

GIMP is completely free and available for Windows, macOS, and Linux.

1. Visit the official website: <https://www.gimp.org/>.
2. Click the ‘Download’ button and select the installer for your operating system.
3. Follow the standard installation prompts.

17.3.2 The Main Interface

When you launch GIMP, the interface is divided into three main areas, as shown in **Figure 17.1**.

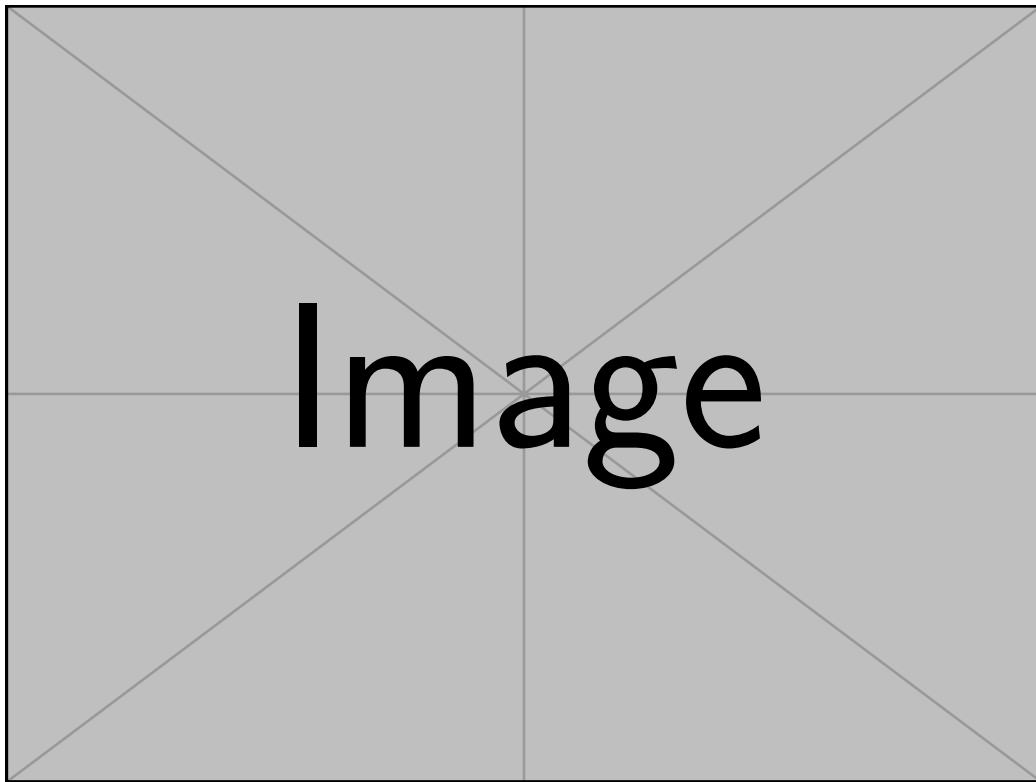


Figure 17.1: GIMP Main Interface showing the Toolbox, Image Canvas, and Layers Dialog.

- **Toolbox (Left):** Contains all your selection, drawing, and transformation tools.
- **Image Canvas (Center):** Your primary workspace.
- **Dockable Dialogs (Right):** This panel houses your Layers, Channels, and Undo History.

17.4 GIMP in a L^AT_EX Workflow

The most important function GIMP serves in your thesis workflow is managing image resolution and file formats before they ever touch your `.tex` files.

17.4.1 Managing Print Resolution (DPI)

A common issue when writing a thesis is that photos taken on a smartphone look great on a screen but print out pixelated and blurry. This is an issue of Dots Per Inch (DPI).

Before exporting an image for L^AT_EX, you must ensure it has a high enough resolution for physical printing:

1. Open your photo in GIMP.
2. Go to ‘Image’ → ‘Print Size’.
3. Ensure the X and Y resolution are set to a minimum of **300 pixels/in**.
4. The dialog will show you the physical size (in inches or millimeters) the image will be when printed at that high resolution.

17.4.2 Cropping for Document Flow

L^AT_EX handles text wrapping and figure placement based strictly on the boundary of the image file. If your photograph has a massive amount of empty floor or ceiling space, L^AT_EX will include that dead space, pushing your text further down the page. Always use GIMP’s Crop tool (‘Shift + C’) to trim your photographs as tightly as possible to the subject matter before exporting.

17.4.3 Exporting Images

Unlike standard programs where you click “Save,” GIMP uses ‘File’ → ‘Save’ to create a `.xcf` project file (which L^AT_EX cannot read). To generate an image for your thesis, you must use ‘File’

→ ‘Export As’.

- **PNG (.png):** Use this format if you have added any text annotations, arrows, or if you have removed the background to make it transparent. PNG prevents the blurry artifacts that JPEG compression causes around sharp text.
- **JPEG (.jpg):** Use this format for standard, illustrative photographs of equipment or field work. It provides good compression for complex colors.

Chapter 18

Dia: Structured Diagramming and Flowcharts

18.1 Introduction

A technical thesis often requires abstract visual representations of your work. Whether you need to map out the logic for a PLC program, draft a block diagram for a hydraulic circuit, or outline the step-by-step workflow of a process, you need a dedicated diagramming tool.

Dia is a free, open-source drawing program designed specifically for structured diagrams. It serves as an excellent, lightweight alternative to commercial software like Microsoft Visio. This chapter will introduce Dia and explain how to create professional, scalable diagrams that integrate perfectly into your L^AT_EX document.

18.2 Key Features of Dia

Dia is not a freehand drawing tool; it is built for precision, structure, and connectivity.

18.2.1 Extensive Shape Libraries

Dia comes pre-loaded with standardized shape libraries for various technical disciplines. You will find dedicated sheets for basic flowcharts, UML (Unified Modeling Language) for software engineering, electrical circuits, pneumatic systems, and network topologies. This ensures your diagrams use industry-standard iconography.

18.2.2 Dynamic Routing and Connectivity

When drafting a complex control logic diagram, the last thing you want is for your lines to break when you move a box. Dia features dynamic connection points. When you anchor a line between two blocks and subsequently reposition one of the blocks, the connecting line automatically stretches and reroutes itself, keeping your system architecture intact.

18.2.3 Snap-to-Grid Alignment

Professional diagrams require perfect alignment. Dia includes a robust grid system and alignment tools that ensure your blocks and flowcharts are perfectly symmetrical, which is critical for the visual appeal of your printed thesis.

18.3 Getting Started with Dia

Dia is incredibly lightweight and easy to get running on any system.

18.3.1 Installation

Dia is available for Windows, macOS, and Linux.

1. For Windows users, the easiest way to download the software is via the installer website:

<http://dia-installer.de/>.

2. Download the latest installer package.
3. Follow the standard installation prompts.

18.3.2 The Main Interface

Upon launching Dia, you will see a straightforward interface that separates your tools from your drawing canvas, as shown in [Figure 18.1](#).

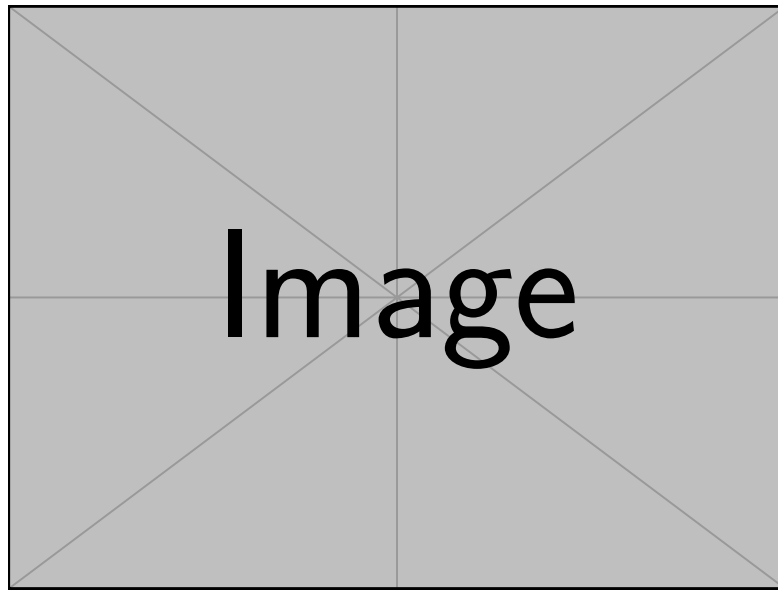


Figure 18.1: Dia Main Interface showing the Toolbox, Shape Selector, and Diagram Canvas.

The primary areas include:

- **Toolbox:** Located on the left, containing basic drawing tools, text insertion, and line connectors.
- **Shape Selector:** Located just below the toolbox, this drop-down menu allows you to switch between different libraries (*e.g.*, from ‘Flowchart’ to ‘Logic’).
- **Canvas:** Your main drawing area, complete with a background grid.

18.4 Dia in a L^AT_EX Workflow

The most crucial aspect of using Dia is how you get your completed diagram out of the software and into your L^AT_EX thesis without losing quality. Because diagrams are made of sharp lines and text, they must be exported as vector graphics.

18.4.1 Exporting as Encapsulated PostScript (EPS)

For a very long time, EPS was the gold standard for L^AT_EX vector graphics.

1. In Dia, go to ‘File’ → ‘Export’.
2. Select **Encapsulated PostScript (*.eps)** from the drop-down format list.
3. This file can be compiled directly into your document. However, note that modern L^AT_EX workflows using `pdflatex` often convert EPS to PDF on the fly during compilation.

18.4.2 Exporting as PDF via Cairo

For modern uAlberta templates that rely on `pdflatex`, exporting directly to a PDF is the safest and cleanest method.

1. Go to ‘File’ → ‘Export’.
2. Choose the **Cairo Portable Document Format (*.pdf)** option.
3. This creates a tightly cropped, infinitely scalable PDF of your diagram that you can insert using the standard `\includegraphics` or the custom `\insertimage` command.

18.4.3 Advanced: Exporting to PGF/TikZ

Just like GNU Octave, Dia has a native export filter for L^AT_EX’s drawing package, **TikZ**. If you want the text in your diagram to perfectly match the font of your thesis (*e.g.*, Times New Roman or Computer Modern), you can export your diagram as a **PGF macro file (*.tex)**.

This generates raw L^AT_EX code. You can then include this diagram in your document using

```
\insertplot{my_diagram.tex}.
```

Note: While exporting to *TikZ* ensures perfect font matching, complex Dia diagrams can generate a massive amount of raw code, which may slightly increase your document compile time. For most students, exporting via the Cairo PDF method strikes the best balance between quality and ease of use.

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Appendix A

Additional Example Figures

Each of the following pages will provide an example of a different figure configuration. In addition to the examples the code that generates the figure will be provided and explanations of what the different parts of the code do will be included. From all of the included information in this Appendix it should be possible to even develop your own figures that potentially suit your needs best.

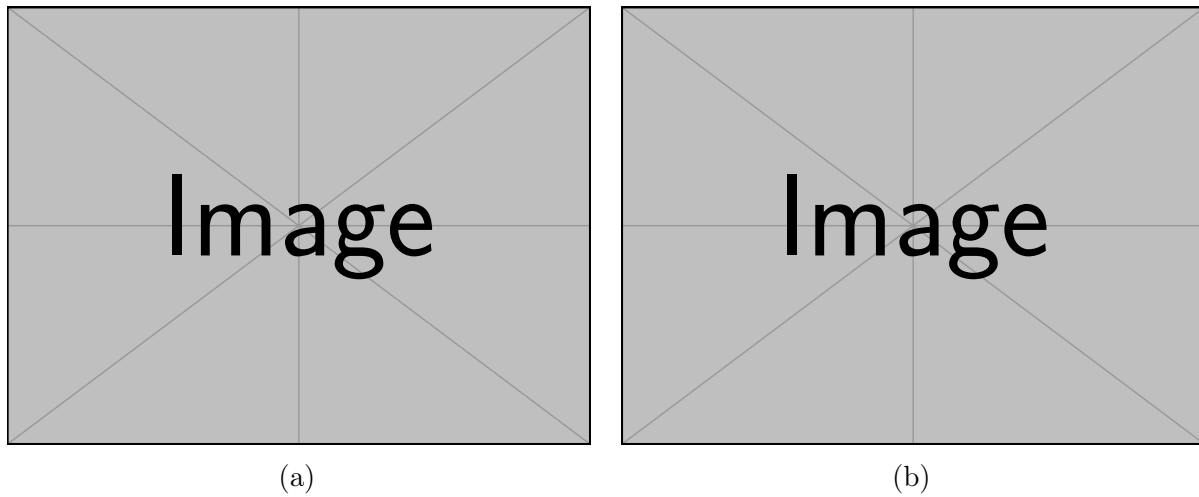
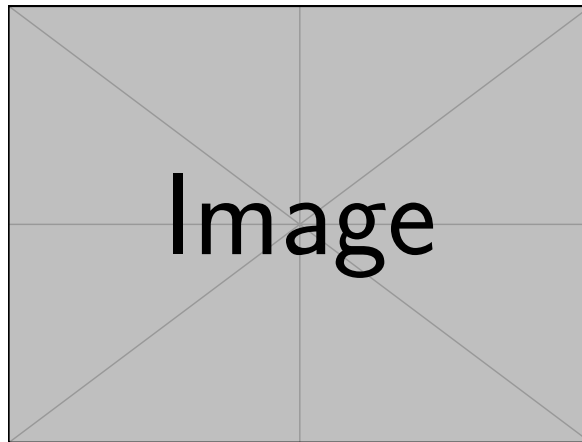


Figure A.1: This is an example of a double image figure.

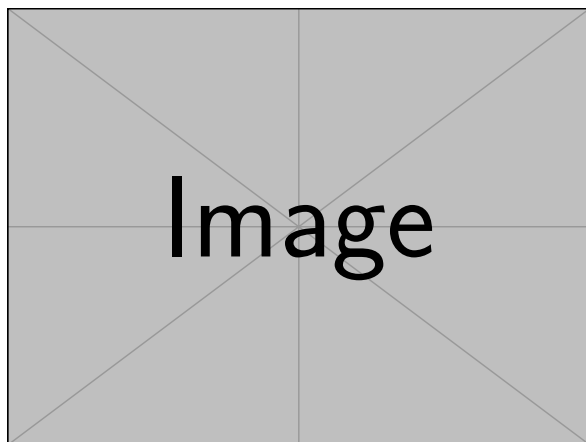
```

\begin{figure}[ht]
  \centering
  \begin{subfigure}{0.45\linewidth}
    \includegraphics[width=\linewidth]{example-image}
    \caption{} % Leave blank for just letter
    \label{fig:doubleImage2:a}
  \end{subfigure}
  ~
  \begin{subfigure}{0.45\linewidth}
    \includegraphics[width=\linewidth]{example-image}
    \caption{} % Leave blank for just letter
    \label{fig:doubleImage2:b}
  \end{subfigure}
  \caption{This is an example of a double image figure.}
  \label{fig:doubleImage2}
\end{figure}

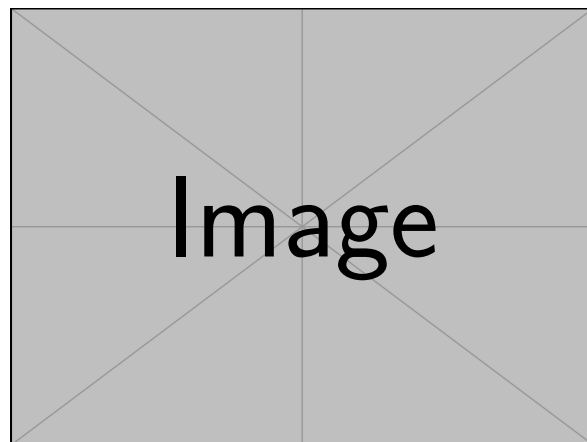
```



(a)



(b)



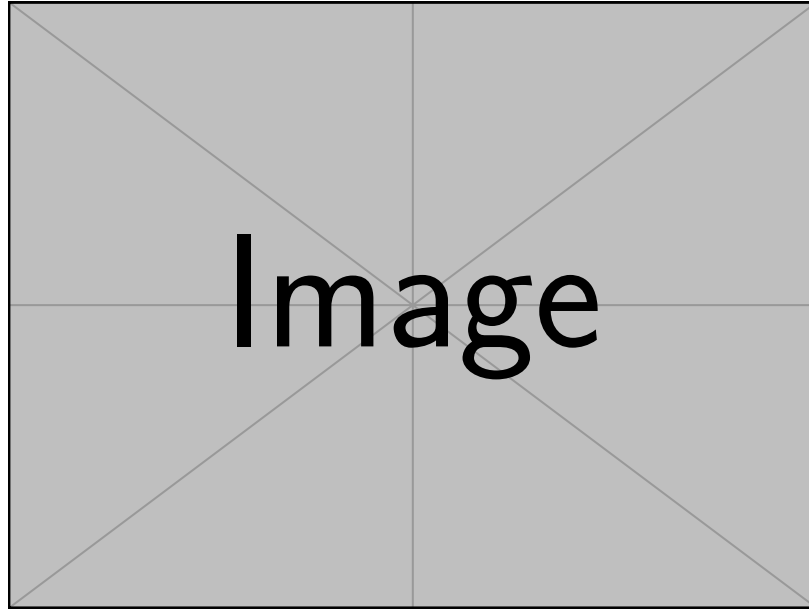
(c)

Figure A.2: This is an example of a triple image figure.

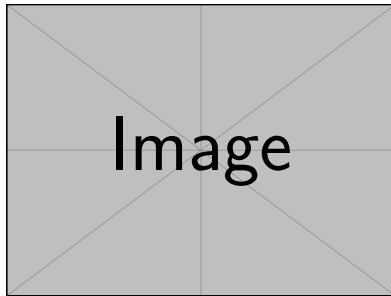
```

\begin{figure}[ht]
\centering
\hspace*{\fill}% Adds space to left of top image (prevents two images from going to top)
\begin{subfigure}{0.45\linewidth}
\includegraphics[width=\linewidth]{example-image}
\caption{} % Leave blank for just letter
\label{fig:tripleImage1:a}
\end{subfigure}
\hspace*{\fill}% Adds space to right of top image (prevents two images from going to top)
\par\vspace{1em}% Adds space between upper and lower images
\begin{subfigure}{0.45\linewidth}
\includegraphics[width=\linewidth]{example-image}
\caption{} % Leave blank for just letter
\label{fig:tripleImage1:b}
\end{subfigure}
~ % Adds space between the two lower figures
\begin{subfigure}{0.45\linewidth}
\includegraphics[width=\linewidth]{example-image}
\caption{} % Leave blank for just letter
\label{fig:tripleImage1:c}
\end{subfigure}
\caption{This is an example of a triple image figure.}
\label{fig:tripleImage1}
\end{figure}

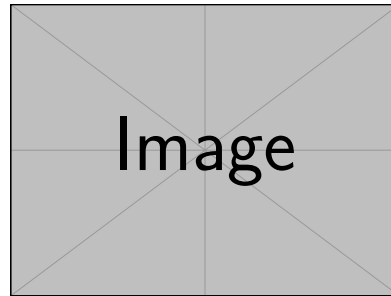
```



(a)



(b)



(c)

Figure A.3: This is a second example of a triple image figure.

```

\begin{figure}[ht]
  \centering
  \hspace*{\fill}% Adds space to left of top image (prevents two images from going to top)
  \begin{subfigure}{0.90\linewidth+1em} % 0.9 = 0.45 + 0.45, and 1em is the width of ~
    \includegraphics[width=\linewidth]{example-image}
    \caption{} % Leave blank for just letter
    \label{fig:tripleImage2:a}
  \end{subfigure}
  \hspace*{\fill}% Adds space to right of top image (prevents two images from going to top)
  \par\vspace{1em}% Adds space between upper and lower images
  \begin{subfigure}{0.45\linewidth}
    \includegraphics[width=\linewidth]{example-image}
    \caption{} % Leave blank for just letter
    \label{fig:tripleImage2:b}
  \end{subfigure}
  ~ % Adds space between the two lower figures
  \begin{subfigure}{0.45\linewidth}
    \includegraphics[width=\linewidth]{example-image}
    \caption{} % Leave blank for just letter
    \label{fig:tripleImage2:c}
  \end{subfigure}
  \caption{This is a second example of a triple image figure.}
  \label{fig:tripleImage2}
\end{figure}

```

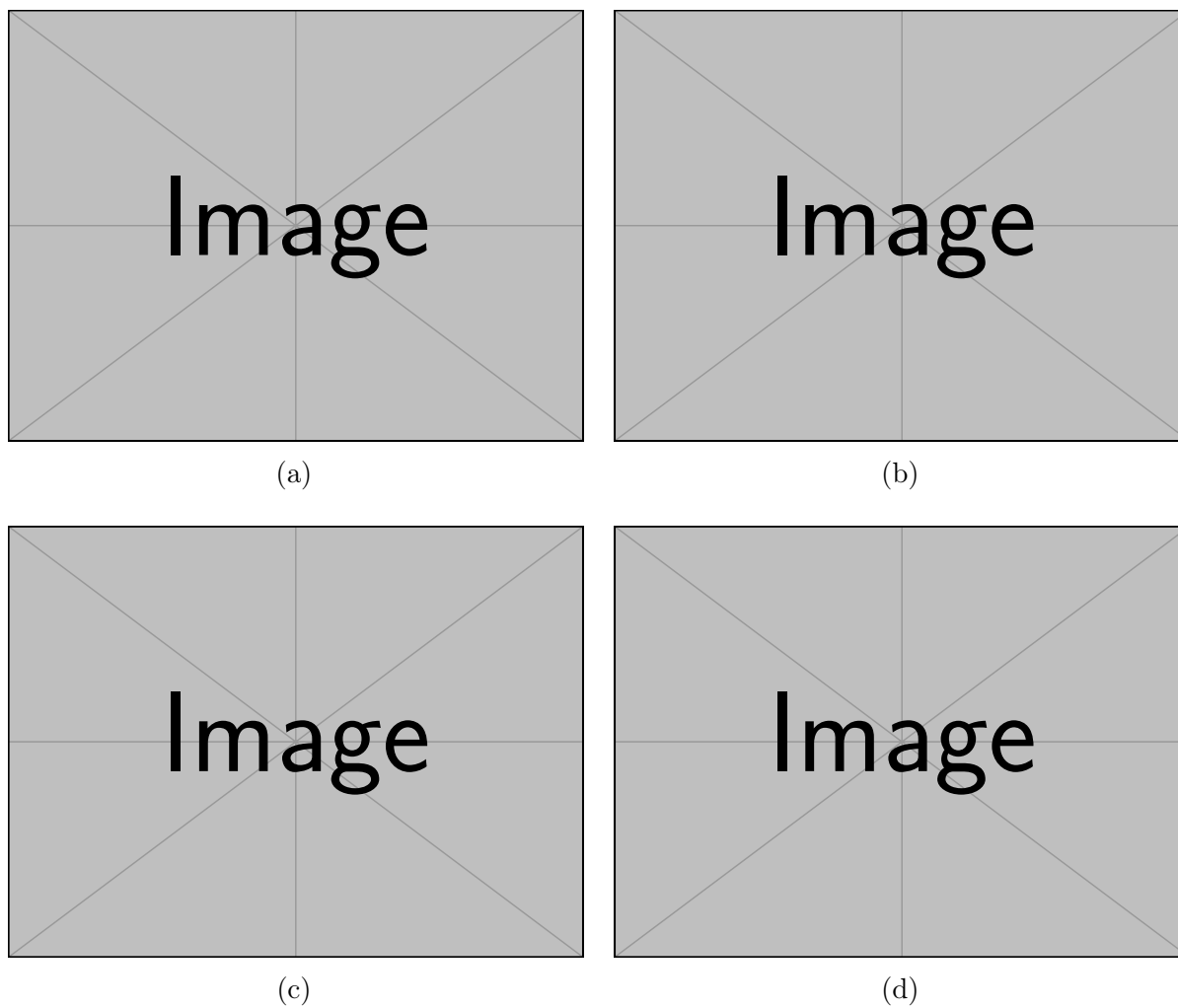


Figure A.4: This is an example of a quad image figure.

```

\begin{figure}[ht]
  \centering
  \begin{subfigure}{0.45\linewidth}
    \includegraphics[width=\linewidth]{example-image}
    \caption{} % Leave blank for just letter
    \label{fig:quadImage:a}
  \end{subfigure}
  ~ % Adds space between the two top figures
  \begin{subfigure}{0.45\linewidth}
    \includegraphics[width=\linewidth]{example-image}
    \caption{} % Leave blank for just letter
    \label{fig:quadImage:b}
  \end{subfigure}
  \par\vspace{1em} % Adds space between upper and lower images
  \begin{subfigure}{0.45\linewidth}
    \includegraphics[width=\linewidth]{example-image}
    \caption{} % Leave blank for just letter
    \label{fig:quadImage:c}
  \end{subfigure}
  ~ % Adds space between the two lower figures
  \begin{subfigure}{0.45\linewidth}
    \includegraphics[width=\linewidth]{example-image}
    \caption{} % Leave blank for just letter
    \label{fig:quadImage:d}
  \end{subfigure}
  \caption{This is an example of a quad image figure.}
  \label{fig:quadImage}
\end{figure}

```

Appendix B

Additional Example Tables

B.1 Section 1

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ultrices. Sed eu leo. In at mauris sit amet tortor bibendum varius. Phasellus justo risus, posuere in, sagittis ac, varius vel, tortor. Quisque id enim. Phasellus consequat, libero pretium nonummy fringilla, tortor lacus vestibulum nunc, ut rhoncus ligula neque id justo. Nullam accumsan euismod nunc. Proin vitae ipsum ac metus dictum tempus. Nam ut wisi. Quisque tortor felis, interdum ac, sodales a, semper a, sem. Curabitur in velit sit amet dui tristique sodales. Vivamus mauris pede, lacinia eget, pellentesque quis, scelerisque eu, est. Aliquam risus. Quisque bibendum pede eu dolor.

B.2 Section 2

Fusce suscipit cursus sem. Vivamus risus mi, egestas ac, imperdiet varius, faucibus quis, leo. Aenean tincidunt. Donec suscipit. Cras id justo quis nibh scelerisque dignissim. Aliquam sagittis elementum dolor. Aenean consectetur justo in pede. Curabitur ullamcorper ligula nec orci. Aliquam purus turpis, aliquam id, ornare vitae, porttitor non, wisi. Maecenas luctus porta lorem. Donec vitae ligula eu ante pretium varius. Proin tortor metus, convallis et, hendrerit non, scelerisque in, urna. Cras quis libero eu ligula bibendum tempor. Vivamus tellus quam, malesuada eu, tempus sed, tempor sed, velit. Donec lacinia auctor libero.

Appendix C

Including Code Listings

This appendix provides guidelines for including code listings in your thesis. Code listings are often used to demonstrate algorithms, data processing scripts, or other relevant programming content. Proper formatting ensures that code is both readable and aesthetically pleasing.

C.1 Using the `listings` Package

The `listings` package is a powerful tool for displaying code in LaTeX. It supports syntax highlighting for a wide variety of programming languages and offers many customization options.

C.1.1 Basic Usage

To include a simple code listing, you can use the following command:

```
\begin{lstlisting}[language=Python]
    # Your code here
    print("Hello, world!")
\end{lstlisting}
```

The `language` option specifies the programming language, which enables syntax highlighting. Replace `Python` with the appropriate language for your code.

C.1.2 Customizing Listings

The `listings` package allows for extensive customization. You can adjust the appearance of your code by setting options such as `frame`, `backgroundcolor`, `keywordstyle`, and more.

Here is an example of how to customize your code listing:

```
\lstset{
    language=Python,
    frame=single,
    backgroundcolor=\color{gray!10},
    keywordstyle=\color{blue}\bfseries,
    commentstyle=\color{green},
    stringstyle=\color{red},
    basicstyle=\ttfamily,
    breaklines=true
}
```

This configuration adds a single-line frame around the code, sets a light gray background, and defines styles for keywords, comments, and strings.

Listing C.1: This is a caption for the inserted code

```
function [outputs] = functionName(inputs)
%{
    This is a Comment Block
    That
    can
    span
    multiple
    lines.
%}

% This is a regular comment
a = 1 + 2 * sin(angle);
b = 'This is a String';
```

Listing C.2: This is a caption for the inserted code

```
#include <iostream>
using namespace std;
/* This function adds two integer values
 * and returns the result
 */
```

```

int sum(int num1, int num2){
    int num3 = num1 + num2; return num3;
}

void main(){
    //Calling the function
    cout << 'The sum is:' << sum(1,99);
}

```

C.2 Advanced Features

C.2.1 Including External Files

The `listings` package allows you to include code from external files. This is particularly useful if you have long code files that you want to reference directly.

```
\lstinputlisting[language=Python]{path/to/your/code.py}
```

Replace `path/to/your/code.py` with the actual path to your file. Or you can use the `\addmedia{./99_Inclusions/}` and `\addcode{Code/}` commands to define the location for code files, then you can just use `\insertcode{filename.ext}` command instead of the full path. You can customize the display in the same way as inline listings.

C.2.2 Handling Special Characters

If your code contains special characters (*e.g.*, #, %, \$), you may need to escape them or use the `literate` option to ensure proper display.

```

\lstset{
    literate={~} {$\sim$}{1}
}

```

This command, for example, replaces the tilde symbol with the appropriate LaTeX command.

C.3 Line Breaks in Long Code Lines

To automatically break long lines of code, use the `breaklines=true` option as shown in the earlier examples. This prevents code from running off the page and maintains readability.

C.4 Conclusion

Including well-formatted code listings in your thesis can enhance the clarity of your work and demonstrate your technical skills. By following the guidelines in this appendix, you can ensure that your code is presented professionally.

Appendix D

Including PDFs

WARNING

While it is possible to have horizontal pages with the page numbers centered on the bottom long edge, I *DO NOT* recommended it. This is because, while it looks okay in a digital format, this is not suitable for printing... this would print page numbers on the side of the page rather than consistently on the bottom or in the heading.

The package that is used to include PDF's is `pdfpages`. This provide the main command `\includepdf` that can be used to include a PDF. Further, this package provides another command, `\includepdfset`, that can be used in the start of the document to pre-set some of the default values. This class file detects the presence of this package and invokes this command as follows:

```
\includepdfset{pages=-,scale=0.85,pagecommand=\thispagestyle{STYLE}}
```

Where `pages=-` defaults to including all pages from the PDF and `scale=0.85` scales the inserted PDF to 85% of its original size, so that the documents fit within the page's margins, headers, and footers.

Further, the option `pagecommand=\cmd {thispagestyle}\mopt {STYLE}` sets the pagestyle

for the PDF pages, where, `STYLE` is replaced with the current `pagestyle`¹ that is in use.

D.1 How to Insert a Portrait PDF

To insert a portrait-oriented PDF into your LaTeX document, you can use the `pdfpages` package, which provides a convenient way to include external PDF files. The following code snippet demonstrates how to include a portrait PDF with the specified options:

```
\includepdf{./99_Inclusions/PDFs/examplePDF}
```

¹To change the `pagestyle` of this one can add or remove the class option `fancyheaders`.

This is an Example PDF that is Portrait

This is the second page

D.2 How to Insert a Landscape PDF

Inserting a landscape-oriented PDF is similarly straightforward using the `pdfpages` package. The code snippet below demonstrates how to include a landscape PDF:

```
\includepdf[landscape=true]{./99_Inclusions/PDFs/landscapePDF}
```

Here, `landscape` sets the orientation to landscape. This configuration ensures that your landscape PDF is correctly oriented and properly sized within your document.

This is an Example PDF that is Landscape

This is the second page

Appendix E

Math Lettering

This page has been intentionally left blank.

Table E.1: Math Mode Greek Letters

Command	Output	Command	Output	Command	Output
<code>\alpha</code>	α	<code>\beta</code>	β	<code>\gamma</code>	γ
<code>\delta</code>	δ	<code>\epsilon</code>	ϵ	<code>\zeta</code>	ζ
<code>\eta</code>	η	<code>\theta</code>	θ	<code>\iota</code>	ι
<code>\kappa</code>	κ	<code>\lambda</code>	λ	<code>\mu</code>	μ
<code>\nu</code>	ν	<code>\xi</code>	ξ	<code>\omicron</code>	$\omicron$
<code>\pi</code>	π	<code>\rho</code>	ρ	<code>\sigma</code>	σ
<code>\tau</code>	τ	<code>\upsilon</code>	υ	<code>\phi</code>	ϕ
<code>\chi</code>	χ	<code>\psi</code>	ψ	<code>\omega</code>	ω
<code>A</code>	A	<code>B</code>	B	<code>\Gamma</code>	Γ
<code>\Delta</code>	Δ	<code>E</code>	E	<code>Z</code>	Z
<code>H</code>	H	<code>\Theta</code>	Θ	<code>I</code>	I
<code>K</code>	K	<code>\Lambda</code>	Λ	<code>M</code>	M
<code>N</code>	N	<code>\Xi</code>	Ξ	<code>O</code>	O
<code>\Pi</code>	Π	<code>P</code>	P	<code>\Sigma</code>	Σ
<code>T</code>	T	<code>\Upsilon</code>	Υ	<code>\Phi</code>	Φ
<code>X</code>	X	<code>\Psi</code>	Ψ	<code>\Omega</code>	Ω

Table E.2: Blackboard Bold Letters

Command	Output	Command	Output	Command	Output
<code>\mathbb{A}</code>	$\mathbb{A}$	<code>\mathbb{B}</code>	$\mathbb{B}$	<code>\mathbb{C}</code>	$\mathbb{C}$
<code>\mathbb{D}</code>	$\mathbb{D}$	<code>\mathbb{E}</code>	$\mathbb{E}$	<code>\mathbb{F}</code>	$\mathbb{F}$
<code>\mathbb{G}</code>	$\mathbb{G}$	<code>\mathbb{H}</code>	$\mathbb{H}$	<code>\mathbb{I}</code>	$\mathbb{I}$
<code>\mathbb{J}</code>	$\mathbb{J}$	<code>\mathbb{K}</code>	$\mathbb{K}$	<code>\mathbb{L}</code>	$\mathbb{L}$
<code>\mathbb{M}</code>	$\mathbb{M}$	<code>\mathbb{N}</code>	$\mathbb{N}$	<code>\mathbb{O}</code>	$\mathbb{O}$
<code>\mathbb{P}</code>	$\mathbb{P}$	<code>\mathbb{Q}</code>	$\mathbb{Q}$	<code>\mathbb{R}</code>	$\mathbb{R}$
<code>\mathbb{S}</code>	$\mathbb{S}$	<code>\mathbb{T}</code>	$\mathbb{T}$	<code>\mathbb{U}</code>	$\mathbb{U}$
<code>\mathbb{V}</code>	$\mathbb{V}$	<code>\mathbb{W}</code>	$\mathbb{W}$	<code>\mathbb{X}</code>	$\mathbb{X}$
<code>\mathbb{Y}</code>	$\mathbb{Y}$	<code>\mathbb{Z}</code>	$\mathbb{Z}$		

Table E.3: Calligraphic Letters

Command	Output	Command	Output	Command	Output
<code>\mathcal\mopt{A}</code>	$\mathcal{A}$	<code>\mathcal\mopt{B}</code>	$\mathcal{B}$	<code>\mathcal\mopt{C}</code>	$\mathcal{C}$
<code>\mathcal\mopt{D}</code>	$\mathcal{D}$	<code>\mathcal\mopt{E}</code>	$\mathcal{E}$	<code>\mathcal\mopt{F}</code>	$\mathcal{F}$
<code>\mathcal\mopt{G}</code>	$\mathcal{G}$	<code>\mathcal\mopt{H}</code>	$\mathcal{H}$	<code>\mathcal\mopt{I}</code>	$\mathcal{I}$
<code>\mathcal\mopt{J}</code>	$\mathcal{J}$	<code>\mathcal\mopt{K}</code>	$\mathcal{K}$	<code>\mathcal\mopt{L}</code>	$\mathcal{L}$
<code>\mathcal\mopt{M}</code>	$\mathcal{M}$	<code>\mathcal\mopt{N}</code>	$\mathcal{N}$	<code>\mathcal\mopt{O}</code>	$\mathcal{O}$
<code>\mathcal\mopt{P}</code>	$\mathcal{P}$	<code>\mathcal\mopt{Q}</code>	$\mathcal{Q}$	<code>\mathcal\mopt{R}</code>	$\mathcal{R}$
<code>\mathcal\mopt{S}</code>	$\mathcal{S}$	<code>\mathcal\mopt{T}</code>	$\mathcal{T}$	<code>\mathcal\mopt{U}</code>	$\mathcal{U}$
<code>\mathcal\mopt{V}</code>	$\mathcal{V}$	<code>\mathcal\mopt{W}</code>	$\mathcal{W}$	<code>\mathcal\mopt{X}</code>	$\mathcal{X}$
<code>\mathcal\mopt{Y}</code>	$\mathcal{Y}$	<code>\mathcal\mopt{Z}</code>	$\mathcal{Z}$		

Table E.4: Fraktur Letters

Command	Output	Command	Output	Command	Output
<code>\mathfrak\mopt{a}</code>	$\mathfrak{a}$	<code>\mathfrak\mopt{b}</code>	$\mathfrak{b}$	<code>\mathfrak\mopt{c}</code>	$\mathfrak{c}$
<code>\mathfrak\mopt{d}</code>	$\mathfrak{d}$	<code>\mathfrak\mopt{e}</code>	$\mathfrak{e}$	<code>\mathfrak\mopt{f}</code>	$\mathfrak{f}$
<code>\mathfrak\mopt{g}</code>	$\mathfrak{g}$	<code>\mathfrak\mopt{h}</code>	$\mathfrak{h}$	<code>\mathfrak\mopt{i}</code>	$\mathfrak{i}$
<code>\mathfrak\mopt{j}</code>	$\mathfrak{j}$	<code>\mathfrak\mopt{k}</code>	$\mathfrak{k}$	<code>\mathfrak\mopt{l}</code>	$\mathfrak{l}$
<code>\mathfrak\mopt{m}</code>	$\mathfrak{m}$	<code>\mathfrak\mopt{n}</code>	$\mathfrak{n}$	<code>\mathfrak\mopt{o}</code>	$\mathfrak{o}$
<code>\mathfrak\mopt{p}</code>	$\mathfrak{p}$	<code>\mathfrak\mopt{q}</code>	$\mathfrak{q}$	<code>\mathfrak\mopt{r}</code>	$\mathfrak{r}$
<code>\mathfrak\mopt{s}</code>	$\mathfrak{s}$	<code>\mathfrak\mopt{t}</code>	$\mathfrak{t}$	<code>\mathfrak\mopt{u}</code>	$\mathfrak{u}$
<code>\mathfrak\mopt{v}</code>	$\mathfrak{v}$	<code>\mathfrak\mopt{w}</code>	$\mathfrak{w}$	<code>\mathfrak\mopt{x}</code>	$\mathfrak{x}$
<code>\mathfrak\mopt{y}</code>	$\mathfrak{y}$	<code>\mathfrak\mopt{z}</code>	$\mathfrak{z}$		
<code>\mathfrak\mopt{A}</code>	$\mathfrak{A}$	<code>\mathfrak\mopt{B}</code>	$\mathfrak{B}$	<code>\mathfrak\mopt{C}</code>	$\mathfrak{C}$
<code>\mathfrak\mopt{D}</code>	$\mathfrak{D}$	<code>\mathfrak\mopt{E}</code>	$\mathfrak{E}$	<code>\mathfrak\mopt{F}</code>	$\mathfrak{F}$
<code>\mathfrak\mopt{G}</code>	$\mathfrak{G}$	<code>\mathfrak\mopt{H}</code>	$\mathfrak{H}$	<code>\mathfrak\mopt{I}</code>	$\mathfrak{I}$
<code>\mathfrak\mopt{J}</code>	$\mathfrak{J}$	<code>\mathfrak\mopt{K}</code>	$\mathfrak{K}$	<code>\mathfrak\mopt{L}</code>	$\mathfrak{L}$
<code>\mathfrak\mopt{M}</code>	$\mathfrak{M}$	<code>\mathfrak\mopt{N}</code>	$\mathfrak{N}$	<code>\mathfrak\mopt{O}</code>	$\mathfrak{O}$
<code>\mathfrak\mopt{P}</code>	$\mathfrak{P}$	<code>\mathfrak\mopt{Q}</code>	$\mathfrak{Q}$	<code>\mathfrak\mopt{R}</code>	$\mathfrak{R}$
<code>\mathfrak\mopt{S}</code>	$\mathfrak{S}$	<code>\mathfrak\mopt{T}</code>	$\mathfrak{T}$	<code>\mathfrak\mopt{U}</code>	$\mathfrak{U}$
<code>\mathfrak\mopt{V}</code>	$\mathfrak{V}$	<code>\mathfrak\mopt{W}</code>	$\mathfrak{W}$	<code>\mathfrak\mopt{X}</code>	$\mathfrak{X}$
<code>\mathfrak\mopt{Y}</code>	$\mathfrak{Y}$	<code>\mathfrak\mopt{Z}</code>	$\mathfrak{Z}$		